

Realizable Nearlinear-Gap Hardness for Small Monotone Formulas

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<https://github.com/Quanyra/realizable-cmmsa-hardness>

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Abstract

For every sufficiently large fixed leaf bound L , we prove randomized polynomial-time many-one NP-hardness of realizable Collection Minimum Monotone Satisfying Assignment with weight gap $L^{1-o(1)}$ and NO satisfaction threshold $o(1)$. The corresponding bounded-advice learning statement follows through Hirahara–Nanashima’s transfer. This establishes the $\epsilon=0$ strengthening asked in Section 7 of their June 23, 2026 revision. We combine their star-projection compilation with Minzer–Zheng’s Grassmann PCP machinery, an altered repetition parameter order, explicit advice-posterior and zoom-out estimates, a maximal-extension threshold ladder, and finite-list completeness repair. Every leaf/advice bound is fixed independently of input length; the polynomial exponent may depend on it. The contribution is the nearlinear exponent in the realizable regime, with earlier realizable hardness explicitly credited below.

Keywords: hardness of approximation; monotone formulas; realizable learning; PCPs.

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1 Problems, conventions, and results

A monotone formula is an AND/OR formula with positive variable leaves; repeated occurrences count separately. $\mathcal{F}[L]$ denotes formulas with at most L leaves. Constants arising from empty branches may be simplified away. For x in $\{0, 1\}^N$ and positive rational weights w_i with $\sum_i w_i=1$, write $w(x) = \sum_{i:x_i=1} w_i$. An instance of $\text{Gap}_\sigma^{\epsilon,\gamma} \mathcal{F}[L]$ -CMMSA is an indexed nonempty explicit list $F_1, \dots, F_M$, those weights, and a rational budget $0 < s \leq 1$. The promises are:

- YES: some x with $w(x) \leq s$ satisfies at least a $1-\epsilon$ fraction of the list.
- NO: every x with $w(x) \leq \sigma s$ satisfies strictly less than a γ fraction.

Indices count with multiplicity. Weights and budgets have polynomial bit encodings and inverse-polynomial lower bounds in the number of variables. This is the normalized-weight restriction of HN Definitions 1.7 and 4.1. The distributional version DMMSA specifies the formula distribution by a sampling circuit. Our final output is an explicit power-of-two list, also represented by its exact uniform sampler. Realizable means $\epsilon=0$.

A randomized polynomial-time many-one reduction maps each input to one output instance, and for each input satisfies the appropriate YES or NO promise with probability at least $2/3$. All

randomness and output lengths are polynomially bounded. The theorem asserts one such uniform reduction on input instances for each fixed L , with no common polynomial exponent asserted across L . Limits as L grows compare this family of separately fixed reductions.

Theorem 1 (Realizable CMMSA). *There are functions σ_L in the positive integers and γ_L in $(0,1)$, with*

$$\frac{\log \sigma_L}{\log L} \longrightarrow 1, \quad \gamma_L \longrightarrow 0, \quad (1.1)$$

such that, for every sufficiently large fixed integer L , $\text{Gap}_{\sigma_L}^{0,\gamma_L} \mathcal{F}[L]$ -CMMSA is NP-hard under randomized polynomial-time many-one reductions. The output weights and budget can be required to share a polynomial-magnitude common denominator.

For completeness we specify the learning model. Fix the universal machine U used by HN. In their Definition 3.11, a distribution on $\{0,1\}^n$ has advice complexity at most a if a function α mapping random strings of length 2^{2n} to strings of length at most a makes $U(\alpha(r); r, n)$ halt on every r , with failure-output probability at most $1/4$ and conditional output statistical distance at most 2^{-2n} from the distribution. The advice can depend on r and need not be computable. This parameter is not the length of a conventional fixed sampler description. For labeled examples use the total encoded sample length in this definition.

$\text{Gap}_{\sigma}^{\epsilon,\gamma} \text{Learn}[a]$ takes unary length and program-budget parameters $1^n, 1^s$ and an example oracle E on $\{0,1\}^n$ times $\{0,1\}$, promised to have sampling advice complexity at most a . Its YES case has a program of description length at most s running in linear time and agreeing with the labels with probability at least $1-\epsilon$. Its NO case bounds the agreement of every program of description length at most σs by $1/2+\gamma/2$. The NO assertion does not impose an efficient running-time restriction on those programs. We use HN's machine and encoding convention; the universal overhead below is the associated constant c_U . Our reduction outputs an explicit sampling circuit, so the hardness also holds with that circuit supplied in place of only oracle access.

Corollary 1 (Realizable learning). *For every sufficiently large fixed integer advice cap a , put*

$$\begin{aligned} L &= a - 2\lceil \log_2(a+1) \rceil - c_U, \\ \sigma_a^{\text{learn}} &= \lfloor 0.49\sigma_L \rfloor, & \gamma_a^{\text{learn}} &= 5\gamma_L. \end{aligned} \quad (1.2)$$

Then $\text{Gap}_{\sigma_a^{\text{learn}}}^{0,\gamma_a^{\text{learn}}} \text{Learn}[a]$ is NP-hard under randomized polynomial-time many-one reductions. For large a the parameters are in their required domains, $\sigma_a^{\text{learn}} = a^{1-o(1)}$, and $\gamma_a^{\text{learn}} = o(1)$. In YES instances the linear-time program has agreement exactly one; in NO instances all programs within the expanded description budget have agreement at most $1/2+\gamma_a^{\text{learn}}/2$.

These are approximation-hardness results under randomized reductions. They give no complexity-class separation, one-way-function construction, linear-factor hardness, or growing-parameter reduction. No assertion about general improper PAC learning or circuit size is made.

2 Relation to previous work

HN revision 1 [3] proves the nearlinear exponent with vanishing nonzero YES error and asks its realizable strengthening separately from its superconstant-parameter question. Hirahara's FOCS 2022 work [2] already proves realizable learning hardness; HN explicitly records the corresponding realizable CMMSA factor L^α for a constant $\alpha>0$, with threshold $1/L^\alpha$. The advance here is the

nearlinear exponent, not the first realizable hardness theorem. The source contracts are the corrected June 23, 2026 revision, not the withdrawn superconstant regime of an earlier version. [3, Definitions 1.2/1.7, Theorems 1.3/1.8, Section 7]; [2].

The MZ [5] higher-query construction gives soundness nearly R^{-m} with $m+1$ queries but its displayed main theorem chooses the alphabet after both positive errors. Our proof changes the construction’s parameter order; substituting an inverse-alphabet completeness error into that main theorem alone would be circular. MZ24’s [6] two-query result, even granting its alphabet-before-completeness abstract formulation, does not provide this higher-arity contract: independent two-query repetition spends two queries per repetition. Likewise a perfect-completeness dictatorship test is not by itself this unconditional star-projection PCP. The targeted literature assessment found no exact subsumer, but does not certify novelty or priority. [5, 6, 1].

OpenAI’s September 23, 2026 preprints present nearby results on Unique Games and perfect-completeness 2-to-1 Games. Their source-hardness and Grassmann/shortcode artifacts are candidate references for individual ingredients. They do not by themselves supply the exact realizable CMMSA star sampler, degree/overlap restrictions, alphabet-soundness tradeoff, encoded runtime, or bounded-advice learning transfer used here. Those bridges remain unverified. Our comparison is a bounded literature screen and does not certify novelty or priority. [8, 7].

Slack variables, AND products, concentration, weight rounding, secret sharing, and the star compilation are established techniques. The new argument is their quantitative composition with the modified repetition, actual advice posterior, conditional rank/zoom-out control, and descending threshold ladder. We retain the source local-decoder statement below for comparison. The enlarged-ambient application used here is the robust $8S$ lemma proved below; the outer hardness, star transport and maximal-pair counting machinery remain cited inputs. The STOC 2026 full text of MZ was not accessible in the assessment; claims about its numerical details are confined to the accessible arXiv v1, also linked by a coauthor.

3 Notation and imported contracts

All vector spaces are over $\mathbb{F}_2$. $\text{Grass}(E, d)$ is the set of d -dimensional linear subspaces of E , with uniform distribution unless stated otherwise. $\begin{bmatrix} n \\ a \end{bmatrix}_2$ is the Gaussian binomial coefficient counting a -subspaces of an n -space. SD is total variation distance (one half the L1 distance). $\text{Zoom}[Q, W]$ is the uniform d -subspace L with $Q \subseteq L \subseteq W$, where $d=2h$. The symbol L for a subspace is local to the Grassmann argument; the scalar L in Theorem 1 always denotes the final leaf bound. Similarly, a denotes advice dimension in that argument and the final advice cap only in Corollary 1. The final asymptotics use base-two log, and $\ln$ is natural log. The parameters m, h, ρ, J are leaf-query count, inner dimension parameter, rational decoding slack, and repetition count. Set $r=10m/\rho$, choosing rational ρ with this value integral. Thus r is a fixed integer bound on advice dimensions and zoom-out codimensions; the local decoder gives $a+c \leq r$.

The following contracts identify the established source results and their parameter ranges. Precise public versions and bibliographic links are in the bibliography. Contract 3 records the scoped source decoder for comparison; our changed-ambient argument uses the separately proved robust $8S$ application. The other contracts remain explicit mathematical imports. None is claimed to be locally Lean-formalized.

Imported contract 1. Outer hardness and game (MZ Theorem 3.1, Section 3.2, Claim 3.2). There is an absolute NO value $s_0 < 1$ for 3-Lin, while its fixed positive YES error may be arbitrarily small. Use the bounded-occurrence form with no pair of variables in more than one

equation. In the smooth repeated game choose J equations independently; in each triple keep all coordinates with probability $1-\beta$, otherwise a uniform single coordinate. Both provers receive the same r random vectors on the retained coordinates, extended by zero to the first question. They answer assignments; acceptance requires agreement on retained variables and satisfaction of the J equations. With fixed NO gap $\eta=1-s_0$, its value is at most $2^{-\Omega(\eta^2 2^{-r\beta J})}$; its honest failure is at most J times the outer YES error. In particular the soundness constant does not depend on that later YES error.

Imported contract 2. Star construction and transport (MZ Section 3.3, Lemmas 3.3–3.4). Keep tuples U of J disjoint equations with no cross-equation pair of variables occurring together in any equation. View U as a $3J$ -space and let H_U be the span of its equation indicator vectors, of dimension J . Labels at $L+H_U$, with $\dim L=2h$ and L transverse to H_U , are linear functions satisfying the equation right-hand sides on H_U ; there are $R=2^{2h}$ labels. Center labels are linear functions on a transverse $2(1-\rho)h$ -subspace K . Two leaf vertices are equivalent when $L+H_U+H'_U=L'+H_U+H'_U$. The imported lemmas give an equivalence relation and unique side-condition-preserving label transport. The test chooses U uniformly, K uniformly transverse, m independent transverse L_i containing K , then a uniform representative from each leaf's equivalence class. It transports each queried label back and tests restriction to K against the center label. This is a weighted star-projection $(m+1)$ -CSP, with alphabets at most R ; all choices are finite and enumerable in $N_{\text{outer}}^{O_m(J)}$. Clique-consistency selection and the collision estimate from Section 5.4.1 preserve the value, up to a vanishing Gaussian-binomial collision loss bounded above by our explicit estimates.

Imported contract 3. Local decoder (MZ Theorem 4.2). For fixed small $\rho>0$ and sufficiently large integral h with integral test dimensions, transverse tables obeying the above side conditions and test density at least $2^{-2(1-1000\rho)hm}$ have dimensions a,c with $a+c\leq 10m/\rho$ such that at least 2^{-6h^2} of the uniform a -subspaces Q admit W containing $Q+H_U$, of codimension c , and a linear g on W respecting H_U , with agreement at least $C=2^{-2(1-1000\rho^2)h}/5$ on the conditioned Grassmann space. This is the source construction's scoped contract. For the changed ambient parameter we derive a robust application with input density at least eight times this threshold below, rather than importing an all- J theorem. Transversality losses are charged explicitly below.

Imported contract 4. Maximal-pair counting (MZ Definition 5.4; MZ24 revision 1 Theorem 5.26). For a total table of linear functions on $\text{Grass}(V, 2h)$, a pair (W, g) is $(B, 1/5)$ -maximal relative to Q if it agrees with the table on at least B of $\text{Zoom}[Q, W]$, and no proper extension W' with a linear extension of g agrees on at least $B/5$ of $\text{Zoom}[Q, W']$. Specialize MZ24 with field $\mathbb{F}_2$ and its $\delta=\rho/m$, fixing its subsidiary constants as in Lemma 5.24 before taking h large. For $\dim V\geq 2h$, $\dim Q\leq 10m/\rho$, $\text{codim } W\leq 10m/\rho$, and $B\geq 2^{-2(1-(\rho/m)^3)h}$, the number of these maximal pairs is at most $B^{-2} 2^{O_{m,\rho}(h)}$. We use the stronger cutoff $B\geq 2^{-2(1-\rho^3)h}$, which implies this one. Thus we require no advice range proportional to J from the broader printed MZ Theorem 5.5. Counting does not assert extension at a single lowered threshold; the threshold ladder below supplies that step.

Imported contract 5. Covering (KMS [4], Definition 4.5, Lemmas 4.6–4.7). For the independent triple-deletion sampler with $2^d \beta\leq 1/8$, basic d -subspace distance is at most $\beta \sqrt{J} 2^{d+4}$. For advice dimension $a<d$, except for at most $\sqrt{\beta} J^{1/4}$ of uniform a -subspaces Q , the distance after conditioning on containing Q is at most $\sqrt{\beta} J^{1/4} 2^{d+5}$. The condition is on the joint sampler,

changing the distribution of V . These covering lemmas are unconditional results proved in KMS Section 8; they do not require its historical combinatorial hypothesis or UGC.

Imported contract 6. Compilation (HN Lemma 4.6). For a star constraint $e=(y;x_1,\dots,x_m)$, let $\pi_{e,i}$ map its leaf alphabet to the center. Its acceptance is $\pi_{e,i}(\text{label}(x_i))=\text{label}(y)$ for every i . Give a vertex u occurrence weight $\lambda(u) = \frac{\mathbb{E}_e[\mathbf{1}[y=u] + \sum_i \mathbf{1}[x_i=u]]}{m+1}$, discarding zero-occurrence vertices. Set $\Lambda = \sum_u \lambda(u)|\Sigma_u|$, $w(z_{u,b})=\lambda(u)/\Lambda$, and $s=1/\Lambda$. If X_e is the set of distinct leaf vertices, let $P_{e,x,b}$ contain labels at x whose projections at every occurrence of x equal b . Output the monotone formula

$$F_e = \bigvee_{b \in \Sigma_y} \left[z_{y,b} \wedge \bigwedge_{x \in X_e} \left(\bigvee_{a \in P_{e,x,b}} z_{x,a} \right) \right]. \quad (3.1)$$

Empty ORs are false. Repeated leaf consistency is imposed in P , not discarded. The formula has at most $(m+1)R$ leaves. Value at least $1-\tau$ gives a weight- s assignment satisfying at least $1-\tau$ of these formulas. CSP value at most ζ implies every assignment of weight at most $\frac{1}{8}(5/8)^{1/(m+1)} \zeta^{-1/(m+1)}$ s satisfies at most $3/4$ of them.

Imported contract 7. Secret sharing and learning (HN Theorem 3.5 and Lemma 5.1). Formula leaf bound L supplies a polynomial-time secret-sharing scheme with total share length at most L . For a circuit-sampled distribution of such formulas, positive inverse-polynomial rational weights and budget, and parameters ϵ in $[0,1]$, γ in $(0,1]$, $\sigma \geq 1$, the randomized transfer preserves ϵ , changes γ to 5γ and σ to 0.49σ , and gives advice at most $L+2\lceil \log_2(L+1) \rceil + c_U$. Its YES guarantee holds for every transfer random string; its NO guarantee fails with only negligible probability. A sufficiently large polynomial length parameter must make all weighted string lengths integral and satisfy the description/soundness bounds. Our common-denominator step supplies this requirement. Integer gap factors are obtained by rounding down.

We now prove the parameter extension and the remaining composition in full. Lemma 1 gives the finite-list repair; Sections 6–10 supply the parameter and reduction arguments for Theorem 1 and Corollary 1. The cited input theorems retain their external-proof status; the scoped decoder in contract 3 is not asserted to have been reproved at its original S threshold.

4 Explicit-list exception lemma

Lemma 1 (Finite-list completeness repair). *Input: $M \geq 1$ monotone formulas $F_1, \dots, F_M$ on N variables, each with at most L leaves; positive rational weights w_i summing to one; positive rational budget $s \leq 1$; integer $\sigma \geq 8$; rational $0 < \Gamma < 1/2$; and rational ϵ with $\epsilon \geq 0$ and $\epsilon \sigma \leq \Gamma/2$. The promises are:*

- *YES: some assignment of weight at most s satisfies at least $1-\epsilon$ of the list, counting indices with multiplicity.*
- *NO: every assignment of weight at most σs satisfies strictly less than Γ of the list.*

Set $\lambda = \sigma s / \Gamma$. Introduce M fresh variables e_j and replace F_j by $F_j \text{ OR } e_j$. Give original variables raw weights w_i and every e_j raw weight λ / M ; divide every weight by $1 + \lambda$. Set

$$t = \frac{s + \lambda \epsilon}{1 + \lambda}, \quad \sigma_{\text{new}} = \lfloor \sigma / 4 \rfloor, \quad \Gamma_{\text{new}} = 2\Gamma. \quad (4.1)$$

The resulting list has at most $L+1$ leaves per formula. In the YES case an assignment of normalized weight at most t satisfies every formula. In the NO case every assignment of normalized weight at most $\sigma_{\text{new}} t$ satisfies strictly less than Γ_{new} of the list.

Proof. In the YES case, take an original witness and enable exactly the exception bits for its failed list indices. Its raw weight is at most $s+\lambda \epsilon$, and it satisfies every augmented formula. No witness or failed-index set is needed to compute the reduction: these choices establish existence.

In the NO case, any augmented assignment of normalized weight at most $\sigma_{\text{new}} t$ has raw weight at most

$$\lfloor \sigma/4 \rfloor (s + \lambda \epsilon) \leq \frac{3\sigma s}{8}. \quad (4.2)$$

Its original-variable part therefore satisfies less than Γ of the original list. Its enabled exception bits occupy at most

$$\frac{3\sigma s/8}{\lambda} = \frac{3\Gamma}{8} \quad (4.3)$$

of list indices. The augmented acceptance fraction is strictly less than $11 \Gamma/8$, hence strictly less than 2Γ . This proves the claimed perfect-completeness gap reduction. Normalization cancels in both budget comparisons, and $t \leq 1$ because $s \leq 1$ and $\epsilon \leq 1$.

The output has $N+M$ variables, M formulas and at most $L+1$ leaves per formula. Construction copies the list, adds M OR gates, and performs rational arithmetic on input rationals. It is polynomial in the actual input and output bit lengths. For fixed σ, Γ , inverse-polynomial lower bounds on w_i and s imply the same for the normalized weights, t , and λ/M . No SAT, optimum, or counting oracle is used. Applying this directly to an exponential implicit distribution would be invalid; sampling precedes it. $\square$

5 Why the repetition parameters are changed

The alphabet of the MZ star construction is $R=2^{2h}$. Its displayed choice $J=2^{100h^2}$, $\beta=\log_2 \log_2 J/J$ has $\beta J=O(\log h)$, so its outer-game estimate alone does not give an exponent proportional to h^2 . We use $J=2^{2^{Ah^2}}$, $\beta=A h^2/J$. This observation concerns only arXiv 2510.23991v1; it is not a disproof of its main theorem or a claim about the uninspected STOC 2026 text. The argument below proves the needed parameter order using the local imported contracts.

6 Parameter extension: posterior and zoom-out argument

This section supplies the parameter modification rather than treating an inverse-alphabet completeness error as part of a black-box theorem. All logarithms here are base two ($\ln$ denotes the natural logarithm). Fix an integer $m \geq 2$ and a positive rational $\rho \leq 1/4000$. Use the local decoding and maximal-pair counting contracts stated above (MZ Theorem 4.2 and the narrower MZ24 Theorem 5.26 specialization). Their parameters r and the constants in their bounds depend on m, ρ . The following lemmas justify the changed ambient application with an explicit constant input margin. Choose h sufficiently large relative to these fixed parameters and a multiple of b_m , the denominator of the selected rational ρ . Then $2(1-\rho)h$ and all test dimensions are integers. All subspaces below are over $\mathbb{F}_2$, U consists of J disjoint triples, and $\dim(U)=3J$. The side-condition space H_U has dimension J .

Lemma 2 (Exact complement identity for star queries). *Let $U = H \oplus Z$, with $\dim H = J$, $\dim Z = 2J$, and $t \leq d \leq 2J$. Draw a uniform t -space K transverse to H , then independent uniform transverse d -spaces $L_i \supseteq K$, for $1 \leq i \leq m$. The law of $(K, H + L_1, \dots, H + L_m)$ equals the law obtained by drawing a uniform complement A of H , a uniform t -space $K \subseteq A$, and independent uniform d -spaces $L'_i \subseteq A$ containing K , and outputting $(K, H + L'_1, \dots, H + L'_m)$.*

Proof. Fix K . Projection to Z expresses it as the graph of a map on a t -space $\overline{K}$. Each possible $D = H + L$ corresponds to a d -space of Z containing $\overline{K}$. Its transverse lifts containing K are exactly the $2^{J(d-t)}$ extensions of that graph map. Consequently the D_i are independent and uniform among these possible spaces. Independently choose a uniform complement A containing K . The map $D_i \mapsto A \cap D_i$ is a bijection to the d -spaces of A containing K , and $H + (A \cap D_i) = D_i$. There are $2^{J(2J-t)}$ complements containing each transverse K , while every complement contains the same Gaussian-binomial number of t -spaces. Thus the uniform incidence law of (K, A) can be sampled in either order, proving the assertion.

The actual leaf table is indexed by $D_i = H + L_i$ and its transported label is restricted to K . The acceptance event is therefore preserved exactly. Random clique representatives can be coupled at identical D_i , with identical transport; alternatively use the already selected clique-consistent table. No condition on intersections among projected leaf increments is needed. This is not an identity of the unobserved raw tuples $(K, L_1, \dots, L_m)$. $\square$

Lemma 3 (Finite matrix-lift pseudorandomness). *Let $E = \mathbb{F}_2^n$, $D = \mathbb{F}_2^d$, $n \geq d$, and $0 \leq r < d$. Suppose $g : \text{Grass}(E, d) \rightarrow \{0, 1\}$ has uniform density at most $e \geq 0$ in every nonempty zoom $\{L : Q \subseteq L \subseteq W, \dim L = d\}$ with $\dim Q + \text{codim}_E W = r$. Define $G(M) = g(\text{im } M)$ for rank- d maps $M : D \rightarrow E$, and $G(M) = 0$ for deficient maps. Then G is basis invariant and its expectation is at most $2e$ on every nonempty affine restriction*

$$\{M : MU = V, XM = Y\}$$

whose nominal budget is $a_0 + b_0 = r$, where U has a_0 columns and X has b_0 rows. Neither family of constraints need be independent. This nominal budget counts vector equations, not ordinary affine-space codimension. Since G is Boolean, the conclusion is the restricted squared- L_2 bound in matrix $(r, 2e)$ -pseudorandomness.

Proof. This reconstructs the finite bridge of MZ Lemma 4.5 and MZ24 Lemmas A.17–A.18, with the strict budget and all affine fibres explicit. For invertible $B : D \rightarrow D$, $\text{im}(MB) = \text{im } M$ and ranks agree, proving basis invariance.

Reduce columns of (U, V) by invertible operations. A dependent column relation violated by V makes the restriction empty; otherwise delete redundant equations and retain $a = \text{rank } U \leq a_0$ independent domain vectors. If their prescribed images are dependent, every satisfying M has a nonzero kernel and $G = 0$. In the remaining case, change domain coordinates to make these the first a columns, denoted V , with independent images. Row operations on (X, Y) similarly delete consistent redundancies (an inconsistent zero row is the empty case). Let $b = \text{rank } X \leq b_0$. Compatibility on the fixed columns is necessary; the remaining uniform restriction has the form

$$M = [V, N], \quad XN = B, \quad k = d - a, \quad a + b \leq r < d.$$

All operations are bijections or deletion of redundant equations; they preserve finite uniform conditional laws and lift values. Empty restrictions have no conditional law and require no assertion. In particular $b < k$.

Reduce (X, B) together so B has $c = \text{rank } B$ independent rows B_1 followed by zero rows; write $X = (X_1; X_0)$ accordingly. The b rows of X are still independent. For $H_0 = \ker X_0$, the map $X_1 : H_0 \rightarrow \mathbb{F}_2^c$ is onto: the combined map (X_1, X_0) is onto, so any target $(y, 0)$ lifts. For uniform free columns $N \in H_0^k$, the target $X_1 N$ is therefore uniform among all c -by- k matrices, with equal fibres. The probability that it has full row rank is

$$\pi(c, k) = \prod_{i=0}^{c-1} (1 - 2^{i-k}) \geq 1 - 2^{-k}(2^c - 1) > \frac{1}{2},$$

using $c \leq b < k$; the empty product at $c = 0$ equals one.

The right action of $\text{GL}(k, 2)$ is transitive on full-row-rank c -by- k targets. The bijection $N \mapsto NB_0$ for such an invertible B_0 preserves $X_0 N = 0$ and the value $G([V, N])$, by right multiplication with $\text{diag}(I_a, B_0)$. Thus its conditional expectation is identical at every full-rank target and equals the expectation conditioned on full target rank. Nonnegativity gives

$$\mathbb{E}[G([V, N]) \mid X_1 N = B_1, X_0 N = 0] \leq \pi(c, k)^{-1} \mathbb{E}[G([V, N]) \mid X_0 N = 0] \leq 2\mathbb{E}_{\text{hom}} G.$$

This is the sole factor two; it concerns the small target matrix, not a large- n rank approximation.

For the homogeneous experiment put $Q = \text{span } V$, $H = \ker X_0$, $W = Q + H$, and $z = \dim(Q \cap H)$. This is ordinary subspace sum; no disjointness is assumed. Its budget satisfies $\dim Q + \text{codim } W \leq a + b - c \leq r$. If no d -space lies between Q and W , the lift is identically zero. Otherwise, for each such L , the map $L \cap H \rightarrow L/Q$ is onto with kernel $Q \cap H$: writing a vector of L as $q + h$ puts h in $L \cap H$. The quotient has dimension k . The free columns yield rank d and image L exactly when their quotient images form an ordered basis. There are exactly

$$|\text{GL}(k, 2)| 2^{zk}$$

choices, independent of L . Conditioning the homogeneous columns on full rank therefore gives the uniform zoom law, and zero on the complement gives

$$\mathbb{E}_{\text{hom}} G = \Pr_{\text{hom}} [\text{rank}[V, N] = d] \mathbb{E}_{L \in \text{Zoom}[Q, W]} g(L) \leq \mathbb{E}_{L \in \text{Zoom}[Q, W]} g(L).$$

No division by the full-rank probability is made.

Finally put $w = \text{codim } W$ and $a' = r - w$. Then $\dim Q \leq a' < d$. Average first over uniform a' -spaces Q' with $Q \subseteq Q' \subseteq W$, then over uniform d -spaces between Q' and W . Both incidence fibre counts are constant: every Q' has the same number of extensions, and every L in the original zoom has the same number of intermediate Q' . The marginal L is consequently uniform on the original zoom. Every refined zoom has exact budget r , so its density is at most e ; averaging proves that bound for the smaller budget as well. Combining the homogeneous estimate with the affine-target estimate proves $\mathbb{E}G \leq 2e$, including all nonempty redundant or dependent cases considered above. $\square$

Lemma 4 (Finite-character spectral bound). *Let $0 \leq s \leq d$, $c = d - s$, and use uniform averages on finite binary matrix spaces, including deficient matrices. Define*

$$\begin{aligned} (\mathcal{T}F)(X) &= \mathbb{E}_{B \in \mathbb{F}_2^{n \times s}} F([X, B]), & X &\in \mathbb{F}_2^{n \times c}, \\ (\mathcal{G}H)(M) &= \mathbb{E}_{R \in \mathbb{F}_2^{d \times c}, \text{rank } R=c} H(MR), \\ (\Phi F)(M) &= \mathbb{E}_{B \in \mathbb{F}_2^{n \times s}, C \in \mathbb{F}_2^{s \times d}, \text{rank } C=s} F(M + BC). \end{aligned}$$

In the last line B, C are independent, B is unrestricted, and only C is conditioned on rank. Suppose $F(MA) = F(M)$ for every $A \in \text{GL}(d, 2)$ and every M . Let F_i be its Fourier projection onto characters indexed by rank- i matrices. Then the $\mathcal{T}F_i$ are mutually orthogonal, and

$$\|\mathcal{T}F_i\|_2^2 = \lambda_i \|F_i\|_2^2, \quad 0 \leq \lambda_i \leq 2^{-is}.$$

In particular the weaker bound used below is valid:

$$\|\mathcal{T}F_i\|_2^2 \leq (2^{-i(s-1)} + 3 \cdot 2^{i-n}) \|F_i\|_2^2.$$

Proof. We reconstruct the finite-character calculation behind MZ Lemma 4.7 and MZ24 Lemmas A.10–A.13 for their actual sampling laws. MZ24 Lemma A.13 states the weaker eigenvalue bound, rather than the exact eigenvalue and energy identities below. Those identities are a direct finite reconstruction for the specified sampling laws; no novelty is claimed. Use $\langle F, H \rangle = \mathbb{E}[F\overline{H}]$ and let J_0 be the inclusion of the first c coordinates. The ordinary adjoint of $\mathcal{T}$ is fixed-coordinate pullback, but for basis-invariant F we may average $M \mapsto MA$ over uniform invertible A to obtain

$$\langle \mathcal{T}F, H \rangle = \mathbb{E}_M F(M) \overline{H(MJ_0)} = \mathbb{E}_{N,A} F(N) \overline{H(NA^{-1}J_0)} = \langle F, \mathcal{G}H \rangle.$$

Each full-column-rank injection $A^{-1}J_0$ has the same number of invertible completions, so its law is the specified uniform law. No invariance of H is required; the identity is not asserted for arbitrary first arguments.

To compute the composition, complete a uniform injection R uniformly to $A = [R, A_2] \in \text{GL}(d, 2)$. At fixed M, A the appended uniform matrix has the law $MA_2 + W$ with W unrestricted. Therefore

$$(\mathcal{G}\mathcal{T}F)(M) = \mathbb{E}_{A,W} F(MA + [0, W]) = \mathbb{E}_{A,W} F(M + [0, W]A^{-1}) = \mathbb{E}_{W,C} F(M + WC) = (\Phi F)(M).$$

Basis invariance justifies the second equality. The last s rows C of A^{-1} are uniform full row rank by constant completion fibres, and W remains independent of C . These statements include empty matrices at $s = 0$ or $c = 0$ and deficient M .

For $S \in \mathbb{F}_2^{n \times d}$ put $\chi_S(M) = (-1)^{\sum_{u,v} S_{uv} M_{uv}}$. Finite character cancellation makes these an orthonormal basis. Translation gives $\Phi\chi_S = \lambda_S \chi_S$, where $\lambda_S = \mathbb{E}_{B,C} \chi_S(BC)$. At fixed C , the exponent pairs B with $SC^\top$. Averaging the unrestricted entries of B cancels to zero unless $SC^\top = 0$, when every term is one. Consequently

$$\lambda_S = \Pr_C[SC^\top = 0].$$

If $\text{rank } S = i$, the rows of C must form an ordered independent s -tuple in the $(d - i)$ -dimensional kernel of S . Counting these tuples yields

$$\lambda_i = \begin{cases} 0, & s > d - i, \\ \prod_{j=0}^{s-1} \frac{2^{d-i} - 2^j}{2^d - 2^j}, & s \leq d - i. \end{cases}$$

The empty product is one. Every nonnegative factor is at most 2^{-i} , so $0 \leq \lambda_i \leq 2^{-is}$. This proves positivity directly and shows that the eigenvalue depends only on rank, not orientation or n .

The identity $\chi_S(MA) = \chi_{SA^\top}(M)$ permutes character indices without changing rank. Thus every rank projection F_i of a basis-invariant F is itself basis invariant, and $\Phi F_i = \lambda_i F_i$; individual characters need not be invariant. The restricted adjoint and composition identities now give

$$\langle \mathcal{T}F_i, \mathcal{T}F_j \rangle = \langle F_i, \Phi F_j \rangle = \lambda_j \langle F_i, F_j \rangle.$$

Orthogonality and the squared-norm formula follow. The unsquared contraction would be $\sqrt{\lambda_i}$, not λ_i . Finally $2^{-is} \leq 2^{-i(s-1)} + 3 \cdot 2^{i-n}$ gives the stated weaker estimate. We retain that weaker form and the conservative ambient conditions in the inverse proof; no constants are reoptimized. $\square$

Lemma 5 (Inverse agreement with explicit ambient bounds). *Fix $m \geq 2$, $0 < \rho \leq 1/4000$, and integral $r = 10m/\rho$. Let h be sufficiently large with integral dimensions $t_0 = 2(1 - \rho)h$, $s_0 = 2\rho h$, $d = 2h$; write*

$$S = 2^{-2(1-1000\rho)hm}, \quad e = 2^{-2(1-1000\rho^2)h}, \quad b = s_0 - 1.$$

For tables of linear functions on the d -spaces and t_0 -spaces of an n -space, suppose the actual test (uniform center and independent uniform containing leaves) has density $\epsilon \geq S$. Under the ambient lower bounds below, some nonempty $\text{Zoom}[Q, W]$ with $\dim Q + \text{codim } W = r$ admits a linear function with agreement greater than e with the leaf table. The constants and the lower cutoffs on h depend only on m, ρ .

Proof. Lemma 3 supplies the matrix lift of an (r, e) -pseudorandom Grassmann indicator as an $(r, 2e)$ -pseudorandom basis-invariant indicator. The strict budget $r < d = 2h$ holds in the regime $h \geq r > 0$ below, and the ambient bounds imply $n \geq d$. Theorem 2 in Appendix A proves the required binary specialization of MZ [5], Theorem 4.6: it bounds the level- i norm at every dyadic $p \geq 4$ by $2^{500i^2p}(2e)^{(p-2)/p}$ for $i \leq r$. The appendix derives this inequality from finite Fourier arguments, with the nominal matrix-restriction interface used here. Lemma 4, with $s = s_0 = 2\rho h$, supplies cross-level orthogonality and the unchanged spectral bound

$$\|\mathcal{T}F^{=i}\|_2^2 \leq (2^{-i(2\rho h-1)} + 3 \cdot 2^{i-n})\|F^{=i}\|_2^2.$$

The matrix and spectral bridges are proved above; the analytic inequality is proved in the appendix. We reconstruct the intervening moment, selection and exponent steps rather than invoke the printed inverse proof without these bounds.

Require $h \geq r$, $e \leq 1/2$, and

$$(2h + 2\rho mh)2^{2h-n} \leq \frac{1}{2}, \quad b \geq 1, \quad n \geq 2h + rb + \log_2 6.$$

The first condition bounds the probability of any column-rank failure in the common-center matrix sampler. Conditioned on full rank, the subspace law is the actual uniform star law; the matrix indicators vanish otherwise. Thus the star density for sets is at most twice the corresponding matrix expectation. For an (r, e) -pseudorandom leaf set with lift F , decompose $\mathcal{T}F = L + H_{\text{high}}$ into levels at most r and above r . Orthogonality and the spectral input give

$$\|H_{\text{high}}\|_2^2 \leq 2^{-(r+1)b} + 3 \cdot 2^{2h-n} \leq 2^{-rb}.$$

Take $\eta = 2^{-(2/3)r\rho h}$. Markov's inequality gives $\Pr[|H_{\text{high}}| > \eta] \leq 2^r 2^{-(2/3)r\rho h}$. On its complement use $|L + H_{\text{high}}|^m \leq 2^{m-1}(|L|^m + \eta^m)$; on the exceptional event use $0 \leq \mathcal{T}F \leq 1$.

Choose a power of two $T \geq \max(4, 4(m+3)/(m\rho))$, and let P be the least power of two at least mT . These depend only on m, ρ . For the lifted center indicator G , whose expectation is at most its Grassmann density β , Hölder contributes $\beta^{1-1/T}$. Probability-space norm monotonicity, contraction of $\mathcal{T}$, and the dyadic norm input at P imply

$$\|L\|_{mT}^m \leq (r+1)^m 2^{500mr^2P} (2e)^{m-2m/P} \leq (r+1)^m 2^{500mr^2P+m} e^{m-2/T}.$$

The last inequality uses $P \geq mT$ and $2e \leq 1$. Combining the splitting and rank factors, the actual matching-star density X for these two sets satisfies

$$X \leq K_{\text{inv}} \beta^{1-1/T} e^{m-2/T} + D_{\text{inv}} 2^{-(20/3)mh},$$

where the deliberately conservative constants are

$$K_{\text{inv}} = 2^{2m}(r+1)^m 2^{500mr^2P}, \quad D_{\text{inv}} = 2^m + 2^{r+1}.$$

In particular we do not use the stronger printed high-degree error or a norm theorem at a possibly nondyadic exponent mT .

For the selection step, a good tuple has its m quotient increments L_i/R in joint direct sum, not merely pairwise trivial intersection. Its total span has dimension $t_0 + ms_0$, so labels agreeing on the center glue to a linear function on that span. Expose ordered leaf bases modulo the center. At increment index j , the previously exposed span has at most 2^{t_0+j} vectors and the allowed sampling denominator is at least 2^{n-1} when $n \geq d+1$. A union bound gives

$$q = \Pr[\text{not jointly direct}] < 2^{t_0+ms_0+1-n} \leq S/2 \quad \text{if } n \geq t_0 + ms_0 + 2 + \log_2(1/S).$$

Thus good accepting tuples have mass at least $\epsilon/2$. This joint-span condition is needed for gluing arbitrary labels in the inverse argument; it is not a loss in Lemma 2's exact quotient-query identity.

Choose a uniform global linear f . Let X_f be the unrestricted star density for which all leaf and center labels match f , and let β_f be its matching-center density. Then

$$\mathbb{E}X_f \geq (\epsilon/2)2^{-(t_0+ms_0)}, \quad \mathbb{E}\beta_f = 2^{-t_0}.$$

Average the affine expression $X_f - (\epsilon/4)2^{-ms_0}\beta_f$ to choose f such that, writing $X = X_f$ and $\beta = \beta_f$,

$$X \geq (\epsilon/4)2^{-ms_0}\beta + (\epsilon/4)2^{-(t_0+ms_0)}, \quad \beta \geq X \geq (\epsilon/4)2^{-(t_0+ms_0)} \geq \frac{1}{4}2^{-2(m+1)h}.$$

These are densities over all actual stars, not densities renormalized by the generic tuples.

Suppose this f 's matching leaf set were (r, e) -pseudorandom. To absorb the analytic error against half of the first selected summand, it suffices to require

$$D_{\text{inv}}2^{-(20/3)mh} \leq (S^2/32)2^{-(t_0+2ms_0)}.$$

Equivalently, the following fixed-parameter lower cutoff suffices:

$$((8/3)m - 2 + (3996m + 2)\rho)h \geq \log_2(32D_{\text{inv}}).$$

Its coefficient is positive. This compares with the full selected signal, including β , rather than with S alone. The selected lower bound and analytic upper bound would now imply

$$1 \leq 8K_{\text{inv}}2^{ms_0}\beta^{-1/T}e^{m-2/T}/\epsilon \leq 8K_{\text{inv}}2^{2/T}2^{-\lambda h},$$

where

$$\lambda = 2000m(\rho - \rho^2) - 2m\rho - 2(m+3)/T \geq m\rho(2000(1 - \rho) - 2.5) > 0.$$

Taking $h > (\log_2(8K_{\text{inv}}) + 2/T)/\lambda$ is a contradiction. By MZ Definition 2.1, failure of (r, e) -pseudorandomness provides $\dim Q + \text{codim } W = r$ and matching density greater than e on a nonempty zoom. The restriction $f|_W$ supplies the asserted agreement. This uses the definition's actual dimensions and gives agreement $> e$, not an unspecified constant times e . All constants above are independent of n . Every ambient requirement is an explicit lower bound, satisfied by $n = 2J$ for the prescribed J and sufficiently large admissible h ; there is no ambient upper bound or required identity involving $\log \log n$. $\square$

Lemma 6 (Robust enlarged-ambient local application). *Fix $m \geq 2$, $0 < \rho \leq 1/4000$ and $r = 10m/\rho$, with integral test dimensions $t = 2(1 - \rho)h$, $d = 2h$. Put*

$$S = 2^{-2(1-1000\rho)hm}, \quad \epsilon' = 2^{-2(1-1000\rho^2)h}, \quad C = \epsilon'/5.$$

For sufficiently large admissible h and $J = 2^{2^{Ah^2}}$, a transverse star test of density at least $8S$ in U , with $\dim U = 3J$ and $\dim H = J$, has fixed a, c with $a + c \leq r$ such that at least 2^{-6h^2} of all uniform a -spaces $Q \subseteq U$ admit a codimension- c space $W \supseteq Q + H$ and a linear function respecting H with agreement at least C on the transverse conditioned Grassmann space. Here $A > 0$ is fixed before h .

Proof. Apply Lemma 5 at $n = 2J$, retaining all its fixed-parameter cutoffs on h and explicit lower bounds on n . It supplies agreement greater than $\epsilon' = e$ at input density at least S . We now give the amplification and side-condition steps.

By Lemma 2, the mean density over complements is at least $8S$. Complements whose density ϵ_A is at least $4S$ have mass at least $4S$, since the mean is at most $4S$ plus that mass. Fix such a complement. Starting from its table, repeatedly apply the inverse argument while the current density is at least $\epsilon_A/2$. Add the resulting zoom to a union X of leaf entries, and independently replace every entry in X by a uniform linear function. The inverse threshold remains valid since $\epsilon_A/2 \geq 2S$.

Here is the preservation event, including its ambient bound. For each candidate (Q, W', g) with dimensions bounded by r , either X has relative measure less than 2^{-2h} in the zoom, or the fraction of the entries in X agreeing with g is at most 2^{1-2h} . Each refreshed entry agrees with a fixed g with probability 2^{-2h} . The Chernoff bound and the lower bound $|X \cap \text{Zoom}[Q, W']| \geq 2^{-2h} 2^{(2h-r)(n-r-2h)}$ in the second case give a per-step failure probability at most

$$F_n = 16(r+1)^2 2^{n(r+1)} \exp(-2^E/12), \quad E = -4h + 2 + (2h - r)(n - r - 2h).$$

Indeed at most $16(r+1)^2 2^{n(r+1)}$ triples are needed: the Gaussian counts for Q, W' contribute at most $16 \cdot 2^{nr}$ and the number of linear g is at most 2^n . The same bound holds conditionally on any preceding history, since each refresh is independent.

Require $2^{1-2h} \leq \epsilon'/2$. Before a new zoom is added, at most this much of its agreement comes from previously refreshed entries; this includes the alternative with relative measure below 2^{-2h} . Its unrefreshed fraction is therefore at least $\epsilon'/2$. The zoom has ambient measure at least $2^{-rn}/4$ by the elementary Gaussian bounds, so X increases by at least $\epsilon' 2^{-rn}/8$. Thus no more than $B_n = \lceil 8 \cdot 2^{rn}/\epsilon' \rceil + 1$ such steps are possible. Choose the parameters so $B_n F_n < 1/2$; a history with all preservation events then exists. This requirement is satisfied for the prescribed J : when $h \geq \max(r+4, 16)$ and $n \geq 8h + 4r + 16$, we have $E \geq hn/2$, whereas $\log B_n + \log(16(r+1)^2 2^{n(r+1)})$ is $O_{m,\rho}(rn + h)$. Every selected function agrees with the original table on at least $\epsilon'/2$ of its zoom.

At termination the acceptance has fallen by at least $\epsilon_A/2$. It can change only if at least one of the m leaf occurrences is in X . Each leaf marginal is uniform, including when leaves repeat, so $\mu(X) \geq \epsilon_A/(2m)$. For a fixed a -space Q , the family of all d -spaces containing Q has measure at most 2^{4h^2-an} . Count distinct Q rather than algorithmic visits: a repeated Q with another zoom still covers only this family. For each distinct useful Q , choose one witnessing W' and assign its codimension c . Let $N_{a,c}$ count the assigned pairs with $\dim Q = a$, let T_a be the number of all a -spaces in the complement, and put $f_{a,c} = N_{a,c}/T_a$. Since $T_a \leq 4 \cdot 2^{an}$,

$$\frac{\epsilon_A}{2m} \leq \mu(X) \leq \sum_{a,c} N_{a,c} 2^{4h^2-an} \leq 4 \cdot 2^{4h^2} \sum_{a,c} f_{a,c}.$$

Thus $\sum_{a,c} f_{a,c} \geq \epsilon_A 2^{-4h^2}/(8m)$. Set these fractions to zero on bad complements and average over all uniform complements. Good complements have mass at least $4S$ and $\epsilon_A \geq 4S$, so the sum of the averaged fractions is at least $2S^2 2^{-4h^2}/m$. A single pigeonhole over at most $(r+1)^2$ dimension pairs gives one fixed (a, c) whose success probability is at least $2S^2 2^{-4h^2}/(m(r+1)^2)$, and hence at least the conservative bound retained below:

$$\frac{2S^2}{m(r+1)^3} 2^{-4h^2}.$$

The pair is fixed after pigeonholing. We do not condition the marginal law on a successful complement: successes form a subset of the full uniform experiment. Its Q marginal is uniform among H -disjoint a -spaces. Extending to uniform $Q \subseteq U$ loses at most 2^{a+1-2J} . The sufficient bounds

$$h^2 \geq 4mh + \log_2(m(r+1)^3) + 1, \quad 2J \geq 5h^2 + r + 2$$

leave at least $2^{-5h^2-1} \geq 2^{-6h^2}$ of all Q . Finally set $W = W' \oplus H$ and extend g by the prescribed side conditions. For fixed A, Q, W' , projection to W/H gives the same uniform quotient law for the containing leaves in W and W' ; the values on H already agree. Hence their agreement events agree and the retained $\epsilon'/2$ is stronger than $C = \epsilon'/5$. This accounts for the uniform- Q correction inside this lemma; subsequent exclusions are applied to its already obtained lucky set. $\square$

Write a for an advice dimension, $0 \leq a \leq r$, and c for a zoom-out codimension, $0 \leq c \leq r$. Write $d=2h$. In the outer sampler each triple retains all three coordinates with probability $1-\beta$, and otherwise retains a uniformly chosen single coordinate. Thus the number of dropped blocks D is $\text{Bin}(J, \beta)$, and $\dim(V)=3J-2D$. Let P be the uniform distribution on a -subspaces Q of U . Let P' be the marginal obtained by first sampling V , then sampling a uniform a -subspace of V .

For $J=2^{2^{Ah^2}}$ and $\beta=A h^2/J$, the advice statistical distance obeys

$$\text{SD}(P, P') \leq \delta_{\text{adv}} := \beta \sqrt{J} 2^{a+4}. \quad (6.1)$$

This is smaller than 2^{-100h^2} for sufficiently large h . Likewise the KMS zoom-in exceptional fraction

$$\delta_{\text{zoom}} := \sqrt{\beta} J^{1/4} \quad (6.2)$$

and the conditional distance $\delta_{\text{zoom}} 2^{d+5}$ are smaller than 2^{-100h^2} , after increasing the fixed lower bound on h . The condition $2^d \beta \leq 1/8$ is satisfied. Increasing A does not cause a circular choice: A depends only on fixed decoding/outer constants, and h is chosen afterward.

Set $T=h^4$ and $\zeta=2^{-30h^2}$. The binomial tail has the explicit bound

$$\Pr[D > T] \leq \left(\frac{eAh^2}{T} \right)^T \leq 2^{-100h^2} \quad (6.3)$$

for sufficiently large h (round T to an integer if needed). Except on a P' -mass at most 2^{-70h^2} of Q , Markov's inequality gives $\Pr[D > T \mid Q] \leq \zeta$. Passing from P' to P adds at most δ_{adv} to this exceptional mass. Further, the set where $P'(Q) < P(Q)/2$ has P -mass at most $2\delta_{\text{adv}}$, by the definition of statistical distance. Intersect these good sets with the KMS good zoom-in set. The excluded P -mass is at most

$$\delta_{\text{zoom}} + 3\delta_{\text{adv}} + 2^{-70h^2} < 2^{-20h^2}. \quad (6.4)$$

Fix a Q outside this exceptional set. For any V with $D \leq T$, Bayes' rule and the Gaussian binomial formula give

$$\begin{aligned} \frac{\Pr[V \mid Q]}{\Pr[V]} &= \frac{\mathbf{1}[Q \subseteq V]}{\begin{bmatrix} \dim V \\ a \end{bmatrix}_2 P'(Q)} \\ &\leq 2 \frac{\begin{bmatrix} 3J \\ a \end{bmatrix}_2}{\begin{bmatrix} 3J-2D \\ a \end{bmatrix}_2} \leq 8 \cdot 2^{2aT}. \end{aligned} \quad (6.5)$$

The last inequality follows by factoring out 2^{2aD} in the product formula; the remaining finite products are bounded by an absolute constant because $3J-2D$ is much larger than a . This is a density bound for the actual posterior, not an assumption that V remains unconditioned.

Now let $W=W(Q)$ be any subspace containing Q , of codimension $c \leq r$. Once Q is fixed, W is fixed before V is sampled. Express W as the kernel of a rank- c matrix. Failure of $\text{codim}_V(W \cap V)=c$ means that some nonzero linear combination of its rows has support entirely outside V . For each of the at most 2^c-1 combinations, select one nonzero coordinate. That coordinate is removed with probability at most β . A union bound therefore gives the unconditional estimate $(2^c-1)\beta$, uniformly in W . Applying the posterior density bound on $D \leq T$, and then the tail bound, gives, for this same Q ,

$$\begin{aligned} \Pr[\text{codim}_V(W \cap V) \neq c \mid Q] &\leq 8 \cdot 2^{2aT} (2^c - 1)\beta + \zeta \\ &\leq 2\zeta. \end{aligned} \quad (6.6)$$

The last inequality holds for large h because J is double exponential in h^2 . The estimate is valid for every $W(Q)$ of the permitted codimension; there is no union over all W , no assumption that W is chosen independently of Q , and no use of an unconditional rank estimate after conditioning.

The KMS distribution conditioned on Q is exactly the posterior needed here. Indeed, the probability a uniform d -subspace L of V contains Q is

$$\mathbf{1}[Q \subseteq V] \frac{\begin{bmatrix} d \\ a \end{bmatrix}_2}{\begin{bmatrix} \dim V \\ a \end{bmatrix}_2}. \quad (6.7)$$

The numerator is independent of V , so conditioning the joint (V, L) sampler on Q subset L induces the same V posterior as sampling Q uniformly inside V . This also explains why an unconditioned outer draw is invalid.

For the zoom-out, put $b=d-a$ and $p_0=2^{-cb}$. On the rank-stable event, the additional probability of L subset W , given V and Q , is

$$p_V = \frac{\begin{bmatrix} \dim V - a - c \\ b \end{bmatrix}_2}{\begin{bmatrix} \dim V - a \\ b \end{bmatrix}_2} = p_0 (1 + O(2^{-J/2})). \quad (6.8)$$

This product estimate is uniform: $\dim(V) \geq J$ always, while a, c, d are $O_{m, \rho}(h)$. Under the uniform U sampler the corresponding probability is at least $p_0/2$ for large h . Conditioning two distributions at distance Δ on an event of probability at least $p_0/2$ changes their distance by at most $O(\Delta/p_0)$. Apply this with the KMS conditional distance $\Delta = \delta_{\text{zoom}} 2^{d+5}$.

Finally compare the posterior mixture with the mixture additionally weighted by p_V . Rank-unstable V have posterior mass at most 2ζ ; their contribution to the normalized weighted mixture is $O(\zeta/p_0)$. On the remaining V , p_V/p_0 differs from one by $O(2^{-J/2})$. Hence for any event concerning L , the difference between these two mixtures is

$$O(2^{-J/2} + \zeta/p_0). \quad (6.9)$$

Both this expression and Δ/p_0 are much smaller than 2^{-10h^2} , since $p_0 \geq 2^{-2rh}$. Therefore agreement with a decoded function on a uniform L satisfying $Q \subseteq L \subseteq W$ transfers to the actual posterior experiment with additive error below 2^{-10h^2} . This supplies the conditional-on- W step with the correct V distribution. It also supplies the rank stability required when the two provers extend their functions.

6.1 Decoder probability is measured in the same joint law

The outer advice consists of shared random vectors in V (MZ Section 3.2), and both provers take the span of the same first a vectors after independently guessing a . Given linear independence, the span is uniform among a -subspaces of V . The probability of dependence is at most $\sum_{i=0}^{a-1} 2^{i-\dim V} \leq 2^{r-J}$. Thus the actual vector-advice law and the ideal experiment V then uniform Q subset V differ in statistical distance by at most 2^{r-J} . This error is negligible compared with every success term below; no alteration of the outer game is needed.

Fix the leaf-label assignment T_1 after the imported clique-consistency selection, with T_2 the center-label assignment. A good first-prover question U is one whose conditional inner-test density reaches the robust threshold $8S$ in the preceding lemma, where $S = 2^{-2(1-1000\rho)hm}$. Write $\Delta = 2^{-2(1-\xi)hm}$. After the already charged clique loss, the surviving density is at least $\Delta/2$. If $\Delta/4 \geq 8S$, the mass of good U is at least $\Delta/4$, since the mean is at most $8S$ plus that mass. The condition $\Delta/S \geq 32$ is exactly $2hm(\xi-1000\rho) \geq 5$. It holds for sufficiently large h with $1000\rho \leq \xi/4$. Thus the good-question mass remains $2^{-O_m(h)}$, with the amplification margin paid explicitly and no additional complement-coupling error.

Fix a good first-prover question U . In the ideal joint experiment the marginal law of Q is P' , and the conditional law of V is exactly the posterior analyzed above. The favorable advice set has P -mass at least $2^{-6h^2} \cdot 2^{-20h^2} \cdot 2^{r+1-2J}$. Passing to P' subtracts δ_{adv} , so its P' -mass is at least 2^{-8h^2} . For each such Q the expected agreement under the actual posterior V is at least $C \cdot 2^{-10h^2}$, hence at least $C/2$. Since agreement is in $[0,1]$, the posterior probability that it is at least $C/4$ is at least $C/4$. Subtracting the rank-failure probability 2ζ leaves at least $C/8$ for large h .

These two probabilities multiply within the actual joint experiment: $P'(\text{good } Q)$ times $\Pr[\text{good } V \mid Q]$. They are never converted to an unconditional V success probability for a fixed Q . In particular, the factor $8 \cdot 2^{2ah^4}$ was used solely to upper-bound a rank-failure event; its reciprocal is not a loss in the decoded success bound. The independent-vector correction subtracts at most 2^{r-J} from the final joint success, again negligible. The threshold-ladder strategy below supplies the maximal-extension step.

6.2 Total tables and transverse subspaces

The source vertices L plus H_U require $L \cap H_U = \{0\}$. For each U , extend the induced table to all d -subspaces by assigning the zero linear function on otherwise undefined entries. The robust enlarged-ambient local lemma above is applied with its proved side-condition/transversality hypothesis. Its resulting agreement on the full conditioned Grassmann space, or equivalently the agreement after charging the exceptional entries, loses a negligible amount quantified here.

If Q intersect $H_U = \{0\}$ and W contains $Q + H_U$ with codimension at most r , a uniform d -subspace L between Q and W fails transversality only if L/Q meets $(H_U + Q)/Q$ nontrivially. A union bound over nonzero vectors of this J -dimensional image bounds this probability by

$$2^{J+d-\dim W+1} \leq 2^{-2J+d+r+1}. \quad (6.10)$$

This follows from $\Pr[z \text{ in a uniform } b\text{-subspace of an } n\text{-space}] = (2^b-1)/(2^n-1)$, and applies with an extra factor at most two if a preceding distribution was already conditioned on transversality. It is negligible compared with C .

For each second-prover question V fix an arbitrary legitimate completion $U_0(V)$ containing V . Define its total table on every d -subspace L of V by $T_1[L+H_{U_0}][L]$ when transverse, and by zero otherwise. This definition depends only on V and the fixed global assignment table. Whenever L is transverse to both H_U and H_{U_0} , the source clique-consistency relation identifies the two values. No equality is asserted on other entries.

Here are explicit charges for those other entries in the actual joint law for a fixed original U . On $D \leq T$, $\dim(V) \geq 3J-2T$. For uniform Q subset V , the probability Q meets $H_{U_0(V)}$ nontrivially is at most $2^{a+J-\dim(V)+1} \leq 2^{-2J+2T+r+1}$. This bound is valid although U_0 is selected after V . Including the tail gives joint probability at most $2^{-100h^2} + 2^{-2J+2T+r+1}$. By Markov, except on P' -mass at most 2^{-69h^2} , its posterior probability for fixed Q is at most ζ . Passing to P adds δ_{adv} . Remove these Q from the good advice set; the total removed advice mass still fits the bound 2^{-20h^2} .

For remaining Q, V with $D \leq T$, rank stability, and Q disjoint from both side-condition spaces, take L uniform between Q and W intersect V . For either side-condition space the same quotient union bound gives nontransversality probability at most

$$2^{J+d-(\dim V-r)+1} \leq 2^{-2J+2T+d+r+1}. \quad (6.11)$$

For H_{U_0} not contained in W intersect V use its intersection with that space; its dimension is at most J , which only improves this bound. Thus replacing either total table by the actually defined transverse values costs at most twice this bound on good V , plus the posterior tail and exceptional probabilities $O(\zeta)$. These are below 2^{-10h^2} . Enlarge the finite lower bound on h to absorb all such additive errors simultaneously. In the subsequent agreement ledger one may use expected agreement $C/2$, good- V agreement $C/4$, and good- V probability $C/8$ as already budgeted. No undefined table entry or unproved identity on nontransverse L is used.

6.3 Explicit threshold ladder for maximal extensions

A single arbitrarily lowered threshold does not guarantee maximality. Instead the second prover guesses j uniformly from $\{0, \dots, r\}$ and uses $B_j = C/(4 \cdot 5^j)$. For a favorable V , begin with $(W(Q) \cap V, g)$ restricted to $W(Q) \cap V$, which has codimension at most r and agreement at least $B_0 = C/4$. At stage j , if the pair is $(B_j, 1/5)$ -maximal, stop. Otherwise Definition 5.4 supplies a proper extension of the pair with agreement at least $B_j/5 = B_{j+1}$. Replace the pair by that extension. Each replacement strictly decreases codimension, so this process stops by stage r , at latest at the whole space, where no proper extension exists.

Thus at one of these $r+1$ thresholds a maximal pair exists whose function extends the first prover's restriction. The second prover enumerates all maximal pairs of codimension at most r at the guessed threshold and chooses uniformly; this is a strategy for a value bound, not a polynomial algorithm used by the reduction. The maximal-pair counting contract bounds each such list by $B_j^{-2} \cdot 2^{O_{m,\rho}(h)} = 2^{O_{m,\rho}(h)}$. Its threshold hypothesis holds: the local agreement is $C = 2^{-2(1-1000\rho^2)h}/5$, so

$$\frac{B_r}{2^{-2(1-\rho^3)h}} = \frac{2^{2(1000\rho^2-\rho^3)h}}{20 \cdot 5^r} \geq 1 \quad (6.12)$$

for sufficiently large h (take $\rho < 1$). Codimension is at most the source local bound $r \leq 10m/\rho$; the advice dimension is also within the narrower MZ24 counting contract. This costs the additional constant factor $1/(r+1)$, with no single-threshold existence assumption. The first prover takes its decoded linear function g on $W(Q)$, which respects the equation side conditions, and extends it uniformly to U . The second prover extends its selected maximal-pair function uniformly to V . For the successful choice, that function extends g restricted to $W(Q) \cap V$. Rank stability gives $\text{codim}_V(W(Q) \cap V) = c$, where $c = \text{codim}_U W(Q) \leq r$. The restriction to V of a uniform extension of g to U is uniform among the 2^c extensions of this common restriction: the quotient map $V/(W(Q) \cap V) \rightarrow U/W(Q)$ is an isomorphism. Thus it agrees with the second prover's chosen extension on V with probability $2^{-c} \geq 2^{-r}$. The first prover's assignment satisfies all outer equations because its extension preserves g on H_U . This proves the claimed random-extension success cost.

7 Completing the modified PCP parameter order

Fix $m \geq 2$ and a desired exponent slack $\xi > 0$; set $\rho \leq \min(\xi/4000, 1/4000)$, choosing it positive and rational and shrinking it further if necessary to meet the imported local theorems. Use the same star-query construction as MZ, with the new J, β above. The following table checks every ambient/repetition-dependent loss used in its decoding argument. The robust local decoding application is proved above. Maximal-zoom-out counting and the cited star/cliue machinery remain imported; the numerical parameter extension is argued here.

Use	Control under the replacement
Ambient dimensions in local decoding	Counting uses $\dim(V) \geq J \geq 2^h$. Complement decoding has ambient $n = 2J$ and its query law is exact. The robust local lemma retains the matrix-rank, spectral, randomization-history and uniform- Q bounds; each ambient error improves with J .
Duplicate clique choices	The source Gaussian-binomial comparison decreases exponentially in J ; for large h the union over m^2 pairs is at most 2^{-J} .
Lucky advice meeting H_U	Its uniform probability is at most 2^{r+1-2J} .
Advice/zoom-in covering	The bounds above and their exceptional Q sets are less than 2^{-20h^2} , below the local decoding lucky mass 2^{-6h^2} .
Zoom-out and rank steps	The posterior calculation above gives error below 2^{-10h^2} , below the decoded agreement $C \geq 2^{-O_{m,\rho}(h)}$.
Number of maximal zoom-outs	The imported bound is $2^{O_{m,\rho}(h)}$ for the relevant agreement and codimension; its constants do not depend on J .
Random extensions	Their agreement cost is at most a factor 2^r , since codimensions are bounded by r .

Write the good-question mass as at least $K_U^{-1} 2^{-B_U h}$ and the maximal-list bound as at most $K_M 2^{B_M h}$, with $K_U, K_M \geq 1$ and $B_U, B_M \geq 0$. These name the bounds just established and imported; their constants depend on m, ρ , not A, J or the later YES error. The displayed decoding strategy, with its previous losses already charged, succeeds before the vector-law correction with probability at least $K^{-1} 2^{-8h^2 - Bh}$, where

$$K = 40K_U K_M (r+1)^3 2^r, \quad B = B_U + B_M + 2.$$

Here $C/8 \geq 2^{-2h}/40$, the three dimension/threshold guesses cost at most $(r+1)^3$, and the extension cost is at most 2^r . For $h \geq \max(1, B + \log_2 K)$ this is at least 2^{-9h^2} . Taking $J \geq 9h^2 + r + 1$, the vector-law error 2^{r-J} leaves at least $2^{-9h^2-1} \geq 2^{-10h^2}$. No exponential-in- J success factor is paid.

The fixed outer NO gap gives success at most $2^{-\kappa\beta J}$, with $\kappa > 0$ depending on the fixed advice size and outer NO gap, not on the later outer YES error. Choose an integer A before h :

$$A > \frac{20}{\kappa}. \quad (7.1)$$

Then the outer upper bound, even including the factor two for legitimate conditioning ensured by the padding below, is at most $2 \cdot 2^{-\kappa A h^2} < 2^{-10h^2}$, contradicting decoded success. All lower bounds on h may depend on this fixed A . Thus the modified PCP has soundness at most $R^{-(1-\xi)m}$, where $R = 2^{2h}$.

Only after fixing m, ξ, h, J, β do we fix any desired positive constant PCP completeness error τ . Select the outer 3-Lin YES error

$$\epsilon_1 \leq \frac{\tau}{100(m+1)J}. \quad (7.2)$$

The starting hardness theorem allows every fixed positive ϵ_1 with an absolute NO gap. In particular, this does not require exactly satisfiable linear equations to be NP-hard. Its reduction exponent may depend on ϵ_1 , which is allowed because every parameter here is a constant independent of the input length.

To account for discarded illegitimate tuples, take sufficiently many disjoint copies of the outer instance so their probability a is at most $\min(\tau/100, 1/4)$. The bound is $O(J^2/N_{\text{outer}})$ with a constant from the bounded occurrence condition. Copies preserve the value and degree bound. The number required depends only on the already fixed parameters; for an input of length n the padded size is $O(n + N_{\min})$, still polynomial. Conditioning on legitimate U multiplies a failure probability by at most $1/(1-a)$. The sampler's marginal for each clique-resampled U'_i is uniform over legitimate U : the initial vertex is uniform, and uniform resampling within its equivalence class preserves that measure. All relevant spaces have the same dimensions and number of transverse extensions.

An honest global labeling satisfies the composed constraint whenever the equations in the original U and all m resampled U'_i are satisfied. Union bounding gives failure at most

$$\frac{(m+1)J\epsilon_1}{1-a} \leq \frac{\tau}{75} < \tau. \quad (7.3)$$

Using all $m+1$ blocks is deliberately conservative; no equality between a single-block bound and the multi-query acceptance event is needed. This establishes the required parameter order: for each fixed m, ξ and each sufficiently large h , every fixed positive τ is available without increasing R . J and the source input-size threshold can be very large, and all are charged as fixed constants.

The construction and enumeration cost is $N_{\text{outer}}^{O_m(J)}$ with constants depending on these choices, plus the starting outer reduction. It is not a polynomial with a uniform exponent when L , h or τ vary with input length. No superconstant-parameter hardness claim is made.

8 From the modified PCP to an explicit realizable list

Use HN's star-projection compilation and its occurrence weights, retaining repeated leaf-variable consistency as in its Lemma 4.6. It gives formulas with at most $(m+1)R$ leaves, a normalized honest budget s , and a constant NO satisfaction bound $3/4$ up to weight σs , where we may safely take the explicit integer

$$\sigma = \left\lfloor \frac{R^{1-1/m}}{16} \right\rfloor, \quad \xi = \frac{1}{m^2}. \quad (8.1)$$

For implementation this floor is computed by integer root comparison; no irrational number is an encoded weight. The constant $1/16$ is smaller than the coefficient in the source's list-decoding bound. For sufficiently large R , $\sigma \geq 8$. Set $q = \lfloor \sqrt{m} \rfloor$ and take an AND of q independently sampled base formulas. Put

$$\Gamma = 2(3/4)^q, \quad \epsilon = \frac{\Gamma}{16\sigma}, \quad \tau = \frac{\epsilon}{4q}. \quad (8.2)$$

The modified PCP above permits this τ after R has been fixed. The AND-product YES failure is at most $q \tau = \epsilon/4$; the NO satisfaction fraction is at most $(3/4)^q = \Gamma/2$. The local leaf bound is $q(m+1)R$.

Let N_0 be the number of monotone variables. Materialize the finite sampling support if necessary; for fixed parameters it has polynomial size in the outer input. Its rational probabilities and occurrence weights have polynomial bit descriptions. A positive atom is a product of a fixed number of inverse-polynomial choices, so positive variable weights and s are inverse-polynomially bounded. Adding rational numbers may enlarge a common denominator, which is handled separately below.

Construct a bounded-random-bit sampler whose distribution has statistical distance at most $\epsilon/8$ from the exact product distribution. One implementation enumerates its S atoms and rounds cumulative rational probabilities to a dyadic grid with at least $\log_2(8S/\epsilon)$ bits. Charge the support enumeration and rational arithmetic. This is polynomial for fixed L ; it is not an oracle for an exponentially large support.

Choose a natural P with $1/\epsilon \leq P$; ϵ is positive. Since ϵ is rational, $P = \lceil 1/\epsilon \rceil$ can be computed by integer division and remainder on its numerator and denominator. Use the conservative count

$$T = 32(N_0 + 11)P^2, \quad M = 2^{\text{clog}_2 T}, \quad (8.3)$$

$$\frac{N_0 \ln 2 + \ln 12}{2(\epsilon/8)^2} \leq T \leq M < 2T.$$

Here $\text{clog}_2 T$ is the least natural exponent whose power of two is at least T . The constructor uses only natural arithmetic. The displayed analytic bound follows from $\ln 2 \leq 1$, $\ln 12 \leq 11$ and $1/\epsilon \leq P$; it is a proved comparison, not a real-number computation required by the sampler. This one count already gives sampling success at least $5/6$, and therefore also the required $2/3$. It is the least qualifying power of two for T , not necessarily for the exact logarithmic threshold. For fixed L the previously chosen positive rational ϵ is fixed, so this choice of P is constant with respect to the outer input, and $M < 64(N_0 + 11)P^2$ bounds the list length. Support enumeration and rational arithmetic costs still have to be charged as above. The formal count and probability bounds do not certify an encoded machine runtime.

Draw M independent formulas from this sampler. Hoeffding's inequality and a union bound over 2^{N_0} assignments imply, with probability at least $2/3$, that all empirical satisfaction fractions differ from their exact product probabilities by at most $\epsilon/4$, including the sampling-distribution error. Hence the list's YES failure is at most $\epsilon/2$, and its NO acceptance is at most $\Gamma/2 + \epsilon/4 < \Gamma$. The exception lemma may therefore use the conservative error budget ϵ . Its condition $\epsilon \sigma \leq \Gamma/2$ holds with a factor-eight margin.

Apply that lemma to this actual list. The output satisfies all constraints in its YES case, with ratio $\lfloor \sigma/4 \rfloor$, threshold 2Γ , and at most $q(m+1)R+1$ leaves per formula. Failure of the randomized reduction occurs only in the sampling event already bounded by $1/3$. The completed list has an

exact uniform sampler using $\log_2(M)$ random bits. In particular, the later learning reduction sees $\epsilon=0$ exactly; it does not inherit the approximate sampler used to construct the list.

9 Rational-weight control and learning transfer

Polynomial bit lengths alone do not make all weights integral multiples of $1/\text{poly}(n)$. We supply a rounding step before any use of fixed-length strings w_i in the learning reduction. Let v_i be the final normalized weights on N' variables and t its YES budget. Choose the least power of two $D \geq 8(N'+1)/t$, set $a_i = \lceil Dv_i \rceil$, $A_0 = \sum_i a_i$, and

$$B_0 = \min\{A_0, \lceil Dt \rceil + N'\}, \quad v'_i = a_i/A_0, \quad t' = B_0/A_0. \quad (9.1)$$

A YES assignment remains feasible, since its integer weight is at most $D t + N'$. Conversely, any assignment with v' -weight at most $(\sigma_{\text{new}}/2)t'$ has original v -weight at most

$$\frac{\sigma_{\text{new}}}{2} \frac{B_0}{D} \leq \frac{\sigma_{\text{new}}}{2} \left(t + \frac{N'+1}{D} \right) \leq \frac{9}{16} \sigma_{\text{new}} t < \sigma_{\text{new}} t. \quad (9.2)$$

Thus the NO acceptance threshold is unchanged if the gap factor is rounded down to $\lfloor \sigma_{\text{new}}/2 \rfloor$. Formula leaves and satisfaction are unchanged. D and A_0 are polynomial numeric quantities because t has an inverse-polynomial lower bound; all new weights and the budget share the polynomial common denominator A_0 . The clipping in B_0 guarantees $t' \leq 1$ and does not weaken either displayed inequality.

Represent the power-of-two list by its exact sampler. Every monotone formula with L leaves admits the source's L -bit-total secret-sharing construction. HN Lemma 5.1 explicitly allows $\epsilon=0$: it keeps that completeness error, multiplies the NO threshold by five, loses a factor 0.49 in the size gap, and uses advice bound

$$L + 2\lceil \log_2(L+1) \rceil + c_U. \quad (9.3)$$

Its length parameter can be chosen as a polynomial multiple of A_0 , so all strings of length v'_i times that parameter have integral lengths. The parameter is also increased by a polynomial factor to meet its stated description and soundness bounds. These adjustments are polynomial in the actual instance size. Perfect completeness holds for every randomness string of this transfer; its additional randomized failure probability is on the NO side. The conservative list count above already gives success at least $5/6$; combine this with the transfer's sufficiently small failure probability to obtain a randomized many-one reduction with success at least $2/3$. Finitely many small inputs can be handled by a fixed truth table, a constant cost depending on the fixed parameters.

10 Final asymptotic parameter choice

For each fixed sufficiently large target leaf bound L , consider integers $m \geq 256$ with $m \leq \sqrt{\log_2 L}$. Let $q = \lfloor \sqrt{m} \rfloor$ and

$$h(L, m) = b_m \left\lfloor \frac{\log_2((L-1)/(q(m+1)))}{2b_m} \right\rfloor, \quad R = 2^{2h}. \quad (10.1)$$

Additionally require $b_m \leq \sqrt{\log_2 L}$. Require $h(L, m)$ to exceed every fixed lower bound used above for that m and $\xi = 1/m^2$, including $\sigma \geq 8$ and all numerical error inequalities. Choose the largest admissible m . Every fixed m is admissible for all sufficiently large L , so $m(L)$ tends to

infinity. The finite selection does not assert a polynomial-time algorithm uniform in L ; for each fixed L it fixes the constants of a uniform polynomial-time reduction on inputs. The same fixed-parameter convention is used in the revised source.

The final leaf bound is at most L by construction. Moreover,

$$\begin{aligned}\log R &= \log L - O(\log(m+1) + b_m) = \log L - o(\log L), \\ \frac{\log \sigma}{\log L} &\longrightarrow 1, \\ \Gamma &= 2(3/4)^{\lfloor \sqrt{m} \rfloor} \longrightarrow 0.\end{aligned}\tag{10.2}$$

All floor operations, the exception lemma, weight rounding and learning transfer lose only constant factors in σ . They therefore preserve $\sigma = L^{1-o(1)}$. For the learning target with advice cap a , choose $L = a - 2\lceil \log_2(a+1) \rceil - c_U$; for large a this is positive and the advice overhead fits within a . Also L/a tends to one, preserving the exponent.

Theorem 1 follows with the explicit parameters

$$\sigma_L = \left\lfloor \frac{\lfloor \sigma/4 \rfloor}{2} \right\rfloor, \quad \gamma_L = 2\Gamma,\tag{10.3}$$

where σ , Γ , m , h and R are selected above. For sufficiently large L these are in the promised ranges. The conservative natural list count above gives sampling success at least $5/6$. Corollary 1 follows from the stated HN transfer with its residual error made at most $1/6$; the combined success is at least $2/3$. The exact sampler and complete repaired YES witness ensure that the learning error parameter is exactly zero. This completes the proofs of Theorem 1 and Corollary 1. $\square$

11 Research and review disclosure

This manuscript was developed through AI-assisted mathematical derivation, source challenge, independent agent mathematical review, dependency audit, and source/significance assessment under Quantyra Research. These are informal AI reviews, not independent human peer review or Lean verification. The source challenger contributed to the argument and is not counted as an independent mathematical verifier. Review roles, findings, and limits are disclosed in the companion review disclosures. Bibliographic versions and imported inputs are recorded in the bibliography.

A Finite binary matrix hypercontractivity

This appendix supplies the analytic inequality used in Lemma 5. We reconstruct the binary finite-matrix argument, including its auxiliary estimates, rather than import a hypercontractive theorem. The underlying method is due to Ellis, Kindler and Lifshitz, and its globalness and dyadic applications to Evra, Kindler and Lifshitz.¹ This is an ordinary mathematical proof, not a claim of Lean verification or novelty. Other upstream PCP, compilation and learning inputs of the manuscript remain as stated.

¹The versions used for comparison are Ellis–Kindler–Lifshitz, *An analogue of Bonami’s lemma for functions on spaces of linear maps, and 2-to-2 games*, arXiv:2209.04243v1, <https://arxiv.org/abs/2209.04243v1>; and Evra–Kindler–Lifshitz, *Polynomial Bogolyubov for special linear groups via tensor rank*, arXiv:2404.00641v2, <https://arxiv.org/abs/2404.00641v2>. The first authors are different. The proof below uses the sufficient constants 6, 196 and an auxiliary weight $6d$, and retains the final constants 100, 103, 200, 500. It does not assert the stronger intermediate displays in those versions.

Theorem 2 (Binary Boolean matrix inequality). *Let $F : \text{Mat}_{n,D}(\mathbb{F}_2) \rightarrow \{0,1\}$ and $0 \leq \delta \leq 1$. Suppose every consistent restriction $MU = V_0$, $XM = Y_0$ of nominal budget r has uniform conditional density at most δ . The nominal budget counts the columns of U plus the rows of X ; dependent and zero equations are allowed. For $0 \leq i \leq r$ and every dyadic integer $p \geq 4$, the rank- i Fourier projection satisfies*

$$\|F^{=i}\|_p \leq 2^{500i^2p} \delta^{1-2/p}.$$

All measures are normalized probabilities. There is no basis-invariance, full-rank-conditioning or aspect-ratio hypothesis.

We prove the theorem after developing its finite ingredients. Dimensions in this appendix may be zero, and all functions other than the final indicator may be complex valued.

A.1 Characters, restrictions and actual derivatives

Put $\mathcal{H}(V, W) = \text{Hom}(V, W)$ for binary vector spaces. Its dual characters are indexed by $Y : W \rightarrow V$: $\chi_Y(M) = (-1)^{\text{Tr}(YM)}$. The matrix trace is in $\mathbb{F}_2$. Its pairing is nondegenerate (test a nonzero matrix entry against the transposed elementary matrix), so character orthogonality gives Fourier inversion and Parseval with $\hat{f}(Y) = \mathbb{E}_M f(M) \chi_Y(M)$. Degree means maximum rank of a nonzero Fourier frequency. For subspaces $A \leq V$, $B \leq W$, write $q_A : V \rightarrow V/A$ and $j_B : B \rightarrow W$ for the canonical maps, and set

$$R_{A,B,T}f(N) = f(T + j_B N q_A), \quad |(A, B)| = \dim A + \text{codim}_W B.$$

Every average on the restriction is uniform on $\mathcal{H}(V/A, B)$. The ordinary filter $P_{A,B}$ keeps frequencies with $A \leq \text{im } Y$ and $\ker Y \leq B$. The hybrid filter $L_{A,B}$ instead keeps

$$A \leq \text{im } Y, \quad Y^{-1}(A) \leq B.$$

Define $D_{A,B,T} = R_{A,B,T} L_{A,B}$ and the influence $I_{A,B,T}(f) = \|D_{A,B,T}f\|_2^2$.

Cyclic trace gives $R_{A,B,T} \chi_Y = \chi_Y(T) \chi_{q_A Y|B}$. For a selected hybrid frequency, $\ker(q_A Y|B) = Y^{-1}(A)$ has dimension $\dim \ker Y + \dim A$. Consequently its rank is $\text{rank } Y - |(A, B)|$. Thus an order- s derivative annihilates levels below s , and for $j \geq s$,

$$D_{A,B,T}(f^{=j}) = (D_{A,B,T}f)^{=j-s}.$$

The below-order assertion is a separate zero case, not truncated natural subtraction. Raw restrictions only decrease rank. Products have additive degree bounds, since $\chi_Y \chi_Z = \chi_{Y+Z}$ and $\text{rank}(Y + Z) \leq \text{rank } Y + \text{rank } Z$.

If $A_2 \leq A_1$ and $B_1 \leq B_2$, a frequency passes the selector for (A_1, B_1) precisely when it passes that for (A_2, B_2) and the induced frequency passes that for $(A_1/A_2, B_1)$. For the forward implication, preimages of A_1 lie in B_1 and supply representatives for every vector of A_1/A_2 . Conversely, quotient representatives and $A_2 \leq \text{im } Y$ give $A_1 \leq \text{im } Y$; if $Yw \in A_1$, choose $z \in B_2$ with $Yz = Yw$ modulo A_2 . Then $w - z \in Y^{-1}(A_2) \leq B_2$, and the second selector puts $w \in B_1$. Character phases therefore prove the exact composition

$$D_{A_1/A_2, B_1, S} D_{A_2, B_2, T} = D_{A_1, B_1, T+j_{B_2} S q_{A_2}}. \quad (\text{A1})$$

All quotient identifications here and below are canonical. In particular $D_{0,W,T}$ is translation, not identity unless $T = 0$. Translation and finite Fubini also prove

$$\mathbb{E}_T \|D_{A,B,T}f\|_2^2 = \|L_{A,B}f\|_2^2 = \sum_{Y \text{ selected}} |\hat{f}(Y)|^2. \quad (\text{A2})$$

At fixed T , frequencies may coalesce, so the last equality is not asserted without the average.

We also need a different derivative. Write $X \preceq Y$ when $\text{rank } Y = \text{rank } X + \text{rank}(Y - X)$. Equivalently, $\text{im } X \leq \text{im } Y$ and X agrees with Y on $Y^{-1}(\text{im } X)$. Indeed rank additivity gives complementary images; conversely the agreement condition forces $\text{im}(Y - X) \leq \text{im } Y$ disjoint from $\text{im } X$, and their sum contains $\text{im } Y$. Let L_X keep $Y \succeq X$ and $D_{X,T} = R_{\text{im } X, \ker X, T} L_X$, with $D_X = D_{X,0}$. For rank- k X , choose splittings making $X = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$. For a fixed induced frequency $Z : \ker X \rightarrow V/\text{im } X$ of rank l , the selected original frequencies are exactly

$$Y = \begin{pmatrix} I + uC & uZ \\ C & Z \end{pmatrix}, \quad u : \text{im } Z \rightarrow \text{im } X, \quad C : \text{im } X \rightarrow \text{im } Z. \quad (\text{A3})$$

Their number is 2^{2kl} . To check this, the off-diagonal blocks of $Y - X$ must factor through Z and land in its image; rank additivity forces the remaining block uC . Conversely choose a lift v with $Zv = C$; then $Y - X = [u; I]Z[v, I]$ has rank l , whereas elimination gives $\text{rank } Y = k + l$. The formula is independent of the lift. Equivalently, the complementary images and kernels from rank additivity force these same blocks. Thus D_X lowers degree by k , with zero for levels below k .

A.2 A weighted fourth-power estimate without contraction

For degree-at-most- d f , put $E = \|f\|_2^2$, $a_X = \sum_{Y \succeq X} |\widehat{f}(Y)|^2$, and $e_X = \|D_X f\|_2^2$. Cauchy–Schwarz on the fibers (A3) gives $e_X \leq 2^{2k(d-k)} a_X$ for $k = \text{rank } X \leq d$; for $k > d$ both are zero. Since $a_X \leq E$, $e_X^2 \leq 2^{4k(d-k)} E a_X$. For a fixed rank- j Y , its rank- k predecessors correspond to rank- k projections on $\text{im } Y$, by $X = QY$. Choosing the image of Q and a complementary kernel counts exactly $\begin{bmatrix} j \\ k \end{bmatrix}_2 2^{k(j-k)}$. The Gaussian product formula gives $\begin{bmatrix} j \\ k \end{bmatrix}_2 \leq 4 \cdot 2^{k(j-k)}$: its denominator product exceeds $1/4$, since its first three factors give $21/64$ and the remaining product is at least $1 - \sum_{a \geq 4} 2^{-a} = 7/8$. The $k = 0$ count is exactly one. Interchanging finite sums now yields

$$\begin{aligned} \sum_X 2^{-6d \text{rank } X} \|D_X f\|_2^4 &\leq E^2 \left(1 + 4 \sum_{k \geq 1} 2^{-6k^2} \right) \\ &\leq \frac{67}{63} E^2 \leq 2E^2. \end{aligned} \quad (\text{W6})$$

Indeed the coefficient at a rank- $j \leq d$ frequency has exponent $-6dk + 4k(d-k) + 2k(j-k) \leq -6k^2$. At $d = 0$ the sum is exactly E^2 .

This argument is necessary: D_X is not a contraction. In fact the weight- $4d$ fourth-power claim in the pinned Ellis–Kindler–Lifshitz version fails for the stated operators. For $V = W = \mathbb{F}_2^d$ and $f = \sum_{Y \in \text{GL}_d} \chi_Y$, write $G_d = |\text{GL}_d|$. Then $E = G_d$, and every rank-one X has $e_X = 2^{4(d-1)} G_{d-1}$ by (A3). There are $(2^d - 1)^2$ such X , and $G_d = (2^d - 1)2^{d-1} G_{d-1}$. Their contribution to the weight- $4d$ sum divided by E^2 is 2^{2d-6} , already 4 at $d = 4$. We use (W6); the later induction has enough slack for it.

A.3 Degree reduction from two convolution classes

We prove the sufficient ordinary-filter inequality

$$\frac{\|f\|_4^4}{162} \leq 2^{6d^2} \|f\|_2^4 + \sum_{(A,B) \neq (0,W)} 2^{7d(\dim A + \text{codim } B)} \|P_{A,B} f\|_4^4. \quad (\text{DR6})$$

Write $a_Y = \widehat{f}(Y)$, $b_Y = |a_Y|$ and $\|f\|_4^4 = \sum_X |\sum_{Y+Z=X} a_Y a_Z|^2$. All contributing Y, Z have rank at most d , and X has rank at most $2d$. Say $M = A \oplus C$ if $M = A + C$ with additive ranks. Let $\mathcal{F}_1(X)$ consist of pairs admitting $Y = A \oplus C$, $Z = B \oplus C$, $X = A \oplus B$. Let $\mathcal{F}_2(X)$ consist of pairs for which

$$\text{im } X \cap \text{im } Y \cap \text{im } Z \neq 0 \quad \text{or} \quad \ker X + \ker Y + \ker Z \neq W.$$

These classes cover all pairs. To prove the assertion outside the second class, put $K_M = \ker M$. One has $(K_Z + K_X) \cap K_Y = K_Z \cap K_Y$: if $w = u + v \in K_Y$ with $u \in K_X$, $v \in K_Z$, then $Zu = Yu = Yv = Xw$ lies in the zero triple image intersection. Define B to equal Z on K_Y and zero on $K_Z + K_X$, and symmetrically A to equal Y on K_Z and zero on $K_Y + K_X$. These are well defined on W , and checking on the three kernels gives $X = A + B$ and $C = Y + A = Z + B$. Their images lie respectively in the pairwise image intersections, so are pairwise disjoint. On K_Z, K_Y, K_X the triples (A, B, C) are $(Y, 0, 0)$, $(0, Z, 0)$, $(0, 0, Y)$; since these kernels span W , the required image sums are onto, proving the direct decompositions.

Put $H_X = \sum_{\mathcal{F}_1(X)} b_Y b_Z$ and $O_X = \sum_{\mathcal{F}_2(X)} a_Y a_Z$. The classes can overlap; partition instead into $\mathcal{F}_2$ and its complement. The convolution is then bounded in absolute value by $H_X + |O_X|$. Consequently its squared sum is at most $2 \sum H_X^2 + 2 \sum |O_X|^2$.

For the first class use independent uniform signs x_A, y_C , indexed by two disjoint copies of all maps of rank at most d , including zero. The real Boolean polynomial

$$\Phi(x, y) = \sum_{A \oplus C} b_{A+C} x_A y_C$$

has degree two. A rank- r map has at most 2^{r^2} direct decompositions: the first summand factors through its kernel quotient and takes values in its image. Thus $\|\Phi\|_2^2 \leq 2^{d^2} \|f\|_2^2$. The elementary Boolean estimate $\|\Phi\|_4^4 \leq 81 \|\Phi\|_2^4$ needs no imported hypercontractivity: for real $G = g + xh$, expansion and Cauchy–Schwarz give $\|G\|_4^4 \leq \alpha^4 + 6\alpha^2\beta^2 + \beta^4 \leq (\alpha^2 + 3\beta^2)^2$, where $\alpha = \|g\|_4$, $\beta = \|h\|_4$. Induction on the sign variables yields $\|G\|_4^2 \leq \sum_S 3^{|S|} \widehat{G}(S)^2$, proving the estimate.

Let $h(A, B) = \sum_C b_{A+C} b_{B+C}$ with both sums direct. Then $H_X \leq \sum_{A \oplus B = X} h(A, B)$. There are at most 2^{4d^2} decompositions of X , so $\sum_X H_X^2 \leq 2^{4d^2} \sum_{A \oplus B} h(A, B)^2$. For $A \neq B$, the coefficient of $x_A x_B$ in Φ^2 is $2h(A, B)$; it dominates the two ordered-pair squares. For a direct pair with $A = B$ one has $A = B = 0$, and $h(0, 0)$ is at most the constant coefficient of Φ^2 . All these coefficients are nonnegative. Parseval therefore gives

$$\sum_X H_X^2 \leq 2^{4d^2} \|\Phi\|_4^4 \leq 81 \cdot 2^{6d^2} \|f\|_2^4. \quad (\text{A4})$$

For the second class preserve complex coefficients and their cancellations. Set $\alpha_k = (-1)^{k+1} 2^{k(k-1)/2}$ for $k \geq 1$. For a t -space, $\mathbf{1}_{t>0} = \sum_{k=1}^t \begin{bmatrix} t \\ k \end{bmatrix}_2 \alpha_k$. This follows by putting $z = -1$ in $\prod_{l=0}^{t-1} (1 + 2^l z) = \sum_k \begin{bmatrix} t \\ k \end{bmatrix}_2 2^{k(k-1)/2} z^k$. The product identity follows inductively from $\begin{bmatrix} t \\ k \end{bmatrix}_2 = \begin{bmatrix} t-1 \\ k \end{bmatrix}_2 + 2^{t-k} \begin{bmatrix} t-1 \\ k-1 \end{bmatrix}_2$, obtained by intersecting subspaces with a fixed hyperplane. Apply the indicator identity to the triple image intersection and to the quotient by the sum of kernels, and use inclusion–exclusion. With $\alpha_{i,0} = \alpha_i$, $\alpha_{0,j} = \alpha_j$, $\alpha_{i,j} = -\alpha_i \alpha_j$ for $i, j > 0$, this gives exactly

$$O_X = \sum_{\substack{(i,j) \neq (0,0) \\ A \leq \text{im } X, \dim A = i \\ B \geq \ker X, \text{codim } B = j}} \alpha_{i,j} (\widehat{P_{A,B} f})^2(X). \quad (\text{A5})$$

Terms with $i > d$ or $j > d$ vanish. For the others, the number of pairs A, B is at most $2^{2d(i+j)}$, because $\text{rank } X \leq 2d$; moreover $|\alpha_{i,j}|^2 \leq 2^{d(i+j)}$. Weighted Cauchy–Schwarz with weight $2^{7d(i+j)}$ has

remaining factor at most $\sum_{i,j \geq 0, (i,j) \neq (0,0)} 2^{-4d(i+j)} < 1$ for $d \geq 1$ (at $d = 1$ it is $31/225$). Summing (A5) over X and using Parseval bounds $\sum_X |O_X|^2$ by the weighted sum in (DR6). Together with (A4), division by 162 proves (DR6). For $d = 0$ it is the constant-function identity with a weakened coefficient. No rank- d bound on the convolution frequency X was used.

A.4 Exact transfers and the fourth-moment induction

Fix $A \leq V$, $B \leq W$. For a frequency Y selected by $P_{A,B}$, put

$$C = Y(B) \cap A, \quad H = B + Y^{-1}(A), \quad R = q_C Y|H.$$

Define $X(b+z) = Yz + C$ for $b \in B$, $z \in Y^{-1}(A)$. Changing the representation changes Yz by $Y(B) \cap A$, so X is well defined, with kernel B and image A/C . Also $Y^{-1}(C) \leq B$, so the hybrid selector accepts Y . The two images $Y(B)/C$ and A/C are disjoint; $R = X + (R - X)$ maps onto these components, so $X \preceq R$.

These properties specify the triple uniquely, and no triple exists for an unselected Y . Indeed suppose a triple $C \leq A$, $H \geq B$, X has those kernel/image conditions, accepts Y by its hybrid selector, and $X \preceq q_C Y|H$. Then $A \leq \text{im } Y$ and $\ker Y \leq B$. Agreement of R with X on $R^{-1}(\text{im } X)$ gives $Y(B) \cap A \leq C$; for $c \in C$, a preimage lies in H and then in $\ker R \leq \ker X = B$, proving the reverse inclusion. For $Yw \in A$, choose $h \in H$ matching Yw modulo C ; the hybrid preimage condition gives $w \in H$. On $Y^{-1}(A)$, $R = X$ and maps onto A/C , whence $H = B + Y^{-1}(A)$. The zero value on B and agreement on this preimage also determine X . On each character the surviving phase is $\chi_Y(T)$, and the final frequency is $q_A Y|B$. Thus, on the actual common domain,

$$R_{A,B,T} P_{A,B} f = \sum_{\substack{C \leq A, H \geq B \\ X: H/B \simeq A/C}} D_X D_{C,H,T} f. \quad (\text{T1})$$

The indexing triples are in bijection with linear maps $\theta : A \rightarrow W/B$: take $C = \ker \theta$, $H/B = \text{im } \theta$ and X the inverse induced isomorphism. Conversely invert X and precompose the quotient of A . Hence their number is exactly 2^{ij} for $i = \dim A$, $j = \text{codim } B$. Holder's inequality for a sum of N functions costs N^3 in fourth powers. Averaging over the base T , whose translation is uniform, therefore gives

$$\|P_{A,B} f\|_4^4 \leq 2^{3ij} \sum_{C,H,X} \mathbb{E}_T \|D_X D_{C,H,T} f\|_4^4.$$

For nonzero ordinary terms $i, j \leq d$. Put $t = \dim C + \text{codim } H + k$, $k = \text{rank } X$. Then $i+j = t+k \leq 2t$, and $7d(i+j) + 3ij \leq 10d(i+j) \leq 24dt$. Each final triple determines the original pair $A = q_C^{-1}(\text{im } X)$, $B = \ker X$ uniquely. Thus (DR6) gives

$$\frac{\|f\|_4^4}{162} \leq 2^{6d^2} \|f\|_2^4 + \sum_{C,H,X:t>0} 2^{24dt} \mathbb{E}_T \|D_X D_{C,H,T} f\|_4^4. \quad (\text{A6})$$

Mixed derivatives with $t > d$ are zero by the two rank-lowering formulas.

We next interchange a later hybrid derivative with D_X . Write $A_1 = \text{im } X$, $B_1 = \ker X$, and fix $A_1 \leq A_2$, $B_2 \leq B_1$. Sum over complements

$$A_2 = A_1 \oplus C, \quad H + B_1 = W, \quad H \cap B_1 = B_2, \quad X' = q_C X|H.$$

Then, with arbitrary actual bases S, T ,

$$D_{A_2/A_1, B_2, T} D_{X, S} f = \sum_{C, H} D_{X'} D_{C, H, S+j_{B_1} T q_{A_1}} f. \quad (\text{T2})$$

Here is a proof of both selector directions and uniqueness. If $X \preceq Y$ passes the left selector, put $Z = Y - X$. Its image is complementary to $\text{im } X$ and $\ker X + \ker Z = W$. On B_1 , the induced Y is the injection of $Z|_{B_1}$ modulo A_1 , and $Z(B_1) = \text{im } Z$. The left hybrid conditions give $C = \text{im } Z \cap A_2$ complementary to A_1 , and $\{w \in B_1 : Zw \in C\} \leq B_2$, in particular $\ker Y \leq B_2$. Take $H = \ker Z + B_2$. It has the required complement properties. If $Yw \in C$, disjoint images force $w \in B_1$ and then $w \in B_2$, so the right hybrid selector holds. On $H = \ker Z + B_2$, the maps X and Z occupy the two complementary components; quotienting by C preserves their disjoint images, proving $X' \preceq q_C Y|_H$.

Conversely suppose a right selector holds. Since $X(H) = \text{im } X$ and $C \leq \text{im } Y$, image inclusion modulo C gives $\text{im } X \leq \text{im } Y$. If $Yw \in \text{im } X$, choose $h \in H$ with $Yh = Yw$: the quotient image condition supplies a match modulo C , and $C \leq Y(H)$ supplies the correction. Then $w - h \in \ker Y \leq H$. Rank-poset agreement modulo C gives $Yw - Xw \in C \cap A_1 = 0$. Thus $X \preceq Y$. Put $Z = Y - X$ again. The hybrid rank drop and $\text{rank } X' = \text{rank } X = k$ give

$$\text{rank}(q_C Z|_H) = \text{rank } Z - \dim C - \text{codim } H.$$

For any Z , the loss under restriction and quotient is at most this sum. Equality forces $\ker Z \leq H$ and $C \leq Z(H)$, hence $Z^{-1}(C) \leq H$. Disjointness from A_1 now forces $C = \text{im } Z \cap A_2$. Moreover $\ker Z + B_2 \leq H$ with equal dimensions: their intersection is $\ker Y$, since $B_2 \leq \ker X$ and $\ker Y \leq H \cap B_1 = B_2$. Rank-nullity gives equality. This proves uniqueness. It also gives the left hybrid conditions, since $Z(B_1) = \text{im } Z$ and $w \in B_1, Zw \in A_2$ implies $w \in H \cap B_1 = B_2$. Finally both sides of (T2) have phase $\chi_Y(S + j_{B_1} T q_{A_1})$ and frequency $q_{A_2} Y|_{B_2}$. This proves the identity without assuming orthogonality of coalescing frequencies.

We prove simultaneously on all finite spaces, by induction on d ,

$$\|f\|_4^4 \leq 2^{100d^2} Q(f), \quad Q(f) = \sum_{A,B} \mathbb{E}_T \|D_{A,B,T} f\|_2^4. \quad (\text{A7})$$

At $d = 0$ only the zero-order derivative survives and equality holds. In (A6) write the initial pair (A_0, B_0) , rank- k map X , $s = \dim A_0 + \text{codim } B_0$, and $t = s + k > 0$. The mixed derivative has degree at most $d - t < d$. Apply the induction hypothesis on its actual smaller space, then (T2) and (A1). A subsequent hybrid derivative has order $u + v \leq d - t$. The two complement choices in (T2) have counts 2^{ku} and 2^{kv} , by writing complements as graphs. Their fourth-power cost is at most $2^{3dk} \leq 2^{6dk}$. The enlarged ordinary order is $s + 2k + u + v \leq d + k$, not necessarily d ; the count only uses the correct mixed bound $s + k + u + v \leq d$. With $A_1/A_0 = \text{im } X$, $B_1 = \ker X$, this yields

$$\mathbb{E}_T \|D_X D_{A_0, B_0, T} f\|_4^4 \leq 2^{100(d-t)^2 + 6dk} \sum_{\substack{A \geq A_0, B \leq B_0 \\ A \cap A_1 = A_0, B + B_1 = B_0}} \mathbb{E}_T \|D_{X|B \bmod A} D_{A,B,T} f\|_2^4. \quad (\text{A8})$$

In (A8), the induced quotient is by A/A_0 inside V/A_0 , with the canonical identification $(V/A_0)/(A/A_0) = V/A$. Each final A, B reconstructs the intermediate pair as $A + A_1$, $B \cap B_1$, so there is no further multiplicity at this stage. For each fixed inner shift, original T plus its canonical embedding is still uniform on the original group; Fubini removes the inner average with no cardinality or rank-conditioning factor.

For completeness we count the remaining reindexing. Fix final A, B, Y of dimensions $a = \dim A, b = \text{codim } B, k = \text{rank } Y$; nonzero terms require $a + b + k \leq d$. Fix initial dimensions $i = \dim A_0, j = \text{codim } B_0$. The exact number of initial triples inducing Y is

$$\begin{bmatrix} a \\ i \end{bmatrix}_2 \begin{bmatrix} b \\ j \end{bmatrix}_2 2^{k(a-i)} 2^{k(b-j)}. \quad (\text{A9})$$

Choose $A_0 \leq A$ and $B_0 \geq B$ first. The image of X is a lift of $\text{im } Y$ to V/A_0 disjoint from A/A_0 , hence a graph with $2^{k(a-i)}$ choices. Its kernel in $B_0/\ker Y$ complements $B/\ker Y$, with $2^{k(b-j)}$ choices. These choices determine X uniquely: project along the chosen kernel, apply Y , then lift its image. This also proves preservation of rank k . The count is at most $2^{d(i+j+k)}$, and we conservatively use $2^{3d(i+j+k)}$. For $t = i + j + k \in [1, d]$, (A6)–(A9) have exponent

$$100(d-t)^2 + 27dt + 6dk \leq 100d^2 - 73dt + 6dk \leq 100d^2 - 63dt - 4dk. \quad (\text{A10})$$

For fixed $k \geq 1$, put $z = 2^{-63d}$ and sum over $i, j \geq 0$: $z^k/(1-z)^2 \leq 2z^k \leq 2^{-31d(k+1)}$. For $k = 0$ omit $i = j = 0$; then $(1-z)^{-2} - 1 \leq 4z \leq 2^{-31d}$. Thus

$$\begin{aligned} \frac{\|f\|_4^4}{162} &\leq 2^{6d^2} \|f\|_2^4 + 2^{100d^2} \sum_{k=0}^d 2^{-31d(k+1)} \\ &\quad \cdot \sum_{\substack{A, B, Y: \text{rank } Y = k \\ |(A, B)| + k > 0}} 2^{-4dk} \mathbb{E}_T \|D_Y D_{A, B, T} f\|_2^4. \end{aligned} \quad (\text{A11})$$

The coefficients satisfy $2^{-31d(k+1)-4dk} = 2^{-31d-29dk} 2^{-6dk}$. Apply (W6), with the common bound d , to each actual $D_{A, B, T} f$, and enlarge sums by nonnegativity. The last line is at most $2^{100d^2+1-31d} Q(f)$. Also $\|f\|_2^4 \leq Q(f)$ by the zero-order term. For $d \geq 1$, $162(2^{-94d^2} + 2^{1-31d}) < 1$, proving (A7). This closes a strict-lower-degree induction; its conclusion was not used at the current degree.

If all influences of a degree-at-most- d function through order d are at most η , (A7) and (A2) imply

$$\|f\|_4^4 \leq 2^{103d^2} \eta \|f\|_2^2. \quad (\text{A12})$$

Indeed an input frequency of rank j is selected by exactly $\sum_{a+b \leq j} \begin{bmatrix} j \\ a \end{bmatrix}_2 \begin{bmatrix} j-a \\ b \end{bmatrix}_2$ hybrid pairs: choose $A \leq \text{im } Y$, then $B \geq Y^{-1}(A)$. For $j \geq 1$ the count is at most $(j+1)2^{j^2} \leq 2^{j^2+2j} \leq 2^{3j^2}$; for $j = 0$ it is one. Replace one factor of each squared influence in $Q(f)$ by η , use (A2), and apply this count. Derivatives above the degree vanish.

A.5 Globalness and both influence conversions

A function is *up-to- r ϵ -global* if every actual affine restriction of order at most r , at every base, has squared L_2 norm at most ϵ . This definition includes the order-zero norm bound. We use this nonvacuous definition also when a quotient space has no restriction of order exactly r .

For a line $U = \langle v \rangle \leq V$, choose $\phi \in V^*$ uniformly with $\phi(v) = 1$ and $w \in W$ uniformly, independently, and define $E_U f(M) = \mathbb{E}_{\phi, w} f(M + w\phi)$. On χ_Y , averaging w keeps only $\phi Y = 0$. If $v \in \text{im } Y$ this is impossible; otherwise exactly $2^{\dim V - \text{rank } Y - 1}$ of the $2^{\dim V - 1}$ functionals qualify. Its multiplier is therefore zero on the order-one hybrid selector and $2^{-\text{rank } Y}$ elsewhere. For a hyperplane $B \leq W$, let ψ be its nonzero defining functional, choose w uniformly with $\psi(w) = 1$ and ϕ uniformly in V^* . The same translation average has multiplier $\Pr[Yw = 0]$, which is zero if $\ker Y \leq B$ and $2^{-\text{rank } Y}$ otherwise. These are direct finite counts; the two distributions remain meaningful in the one-dimensional and singleton boundary cases whenever the line or hyperplane exists. For either operator E , with its hybrid selector L , one has

$$L(f^{\neq j}) = f^{\neq j} - 2^j E(f^{\neq j}). \quad (\text{A13})$$

For $j \geq 1$ set $P_j = (I - 2^j E)(I - 2^{j-1} E)$. On both ranks $j, j-1$ it equals L . A raw order-one restriction loses exactly one rank on selected frequencies and zero otherwise. Hence only those two input ranks can reach output rank $j-1$, and

$$(R_{U, T} P_j f)^{\neq j-1} = D_{U, T}(f^{\neq j}). \quad (\text{A14})$$

The selected rank-zero part is zero when $j = 1$; no negative rank is identified with rank zero. This identity holds for arbitrary f and every T .

Translations preserve globalness at all bases. Minkowski implies the same for any convex mixture of translations, in particular E, E^2 . The coefficient sum of P_j is $(1 + 2^j)(1 + 2^{j-1}) \leq 2^{2j+1}$. Thus, if f is up-to- j ϵ -global, the actual witness $f' = R_{U,T}P_j f$ is up-to- $(j-1)$ $4 \cdot 2^{4j} \epsilon$ -global and has level $j-1$ given by (A14). Indeed composing a further restriction with $R_{U,T}$ adds one to its order and adds its canonical embedded base to T , preserving the normalized measure. This works for both a domain line and codomain hyperplane, and uses the requested base T , not only zero.

For a derivative of order $k \leq d$, peel off a line in its nonzero domain constraint, or otherwise a hyperplane containing its codomain constraint. Apply this witness at the first base T and subsequently at zero. Formula (A1) preserves the specified total base. After k steps, Parseval for the remaining level gives

$$I_{A,B,T}(f^{=d}) \leq 2^{4kd-2k^2+4k} \epsilon \leq 2^{10d^2} \epsilon. \quad (\text{A15})$$

Here f was up-to- d global but had no degree assumption. The product of losses is $\prod_{j=d-k+1}^d 4 \cdot 2^{4j}$; its exponent is at most $2d^2 + 4d$ for $d \geq 1$. At $k = 0$ use the whole-space Fourier projection bound, and at $d = 0$ this is the only case. Thus all orders, not just order d , are covered. For a full degree-at-most- d function, the images of its levels under a fixed hybrid derivative are orthogonal, having distinct ranks $i - k$. Applying (A15) to each level and summing gives

$$I_{A,B,T}(f) \leq 2^{11d^2} \epsilon \quad (k \leq d), \quad (\text{A16})$$

since $(d+1)2^{10d^2} \leq 2^{11d^2}$ for $d \geq 1$. The $d = 0$ assertion is direct. This uses orthogonality between levels, not between colliding individual frequencies.

We also prove the reverse implication: if a degree-at-most- d function has all influences through order d at most ϵ , then for every natural r it is up-to- r $2^{10dr} \epsilon$ -global. First note that level projections inherit the influence bound, because they are orthogonal projections of the actual reduced derivative. Order-one derivatives inherit all influences through $d-1$ by (A1).

The only additional measure calculation is the following. Suppose f is up-to- s η -global. Restrict $E_U f$ first at a line $U = \langle v \rangle$ and then at further order at most s . Write the parent variation as $R \in \mathcal{H}(V/A, B)$ with $U \leq A$ and $\dim A + \text{codim } B \leq s+1$. Jensen bounds its squared norm by $\mathbb{E}_{R,\phi,w} |f(M_0 + R + w\phi)|^2$. Condition on $A' = \ker(\phi|_A)$ and $B' = B + \langle w \rangle$. The restriction of ϕ to A is fixed by A' and $\phi(v) = 1$; outside A its coordinates remain independently uniform. In coordinates on V/A' starting with v , R has first column zero and other columns independently uniform in B . If $B' = B$, w is uniform in B , and the sum is uniform on $\mathcal{H}(V/A', B)$. If $\dim B' = \dim B + 1$, w is uniform in $B' \setminus B$; every other column is an independent uniform element of B plus an independent uniform bit times w , hence uniform in B' . Thus the resulting map is uniform conditional on its v -column being outside B , an event of density exactly $1/2$. The unconditional restriction orders are at most s and $s-1$, respectively. Nonnegative expectation under this conditioning costs at most two, so the desired squared norm is at most 2η . When $s = 0$ the expanded case is impossible ($B = W$), so no negative-order hypothesis is used. Arbitrary bases are absorbed by translation. Codomain hyperplanes follow by transposing the spaces: restrictions (A, B) become $(B^\perp, A^\perp)$ with the same order and uniform law.

For homogeneous f of degree $d \geq 1$, (A13) reads $f = L_U f + 2^d E_U f$. If f is global through $r-1$ with parameter η_2 and every order-one derivative is global through $r-1$ with parameter η_1 , the preceding calculation and squared triangle inequality bound each order- r restriction by

$$2\eta_1 + 4 \cdot 2^{2d} \eta_2. \quad (\text{A17})$$

Induct first on d simultaneously for all spaces and all r , then on r . The $d = 0$ and $r = 0$ cases follow from order-zero influence. For $d, r \geq 1$ split into rank levels. Lower levels have global bounds $2^{10ri}\epsilon$ by outer induction. The top level has the $r - 1$ bound $2^{10d(r-1)}\epsilon$ by inner induction, and its order-one derivatives have the bound $2^{10(d-1)(r-1)}\epsilon$ by outer induction. Relative to $K = 2^{10dr}\epsilon$, (A17) costs at most

$$2 \cdot 2^{-10(r+d-1)} + 4 \cdot 2^{-8d} \leq 9/512 < 1/4.$$

Smaller-order top restrictions already satisfy the same $K/4$ bound. For all lower levels together the squared triangle and geometric bound is at most $\epsilon(\sum_{i < d} 2^{5ri})^2 \leq K/961$. Combining the top and lower sums costs at most $K/2 + 2K/961 < K$. This proves

$$\text{influences through } d \leq \epsilon \implies \text{globalness through } r \leq 2^{10dr}\epsilon \text{ for all } r \geq 0. \quad (\text{A18})$$

If exact-order- r restrictions do not exist, use the smaller-order induction result, not a choice from an empty family. If a level is unavailable it is zero. The case $\epsilon = 0$ follows directly from order-zero norm, without division.

A.6 Square-globalness and all dyadic moments

Suppose f has degree at most d and is up-to- d ϵ -global. Combining (A16) with (A12) gives

$$\|f\|_4^4 \leq 2^{114d^2}\epsilon\|f\|_2^2. \quad (\text{A19})$$

Combining (A16) with (A18) at $r = 3d$ gives up-to- $3d$ globalness parameter $B = 2^{41d^2}\epsilon$. Every raw restriction h of order at most $2d$ still has degree at most d and is up-to- d B -global, by composition of restrictions. Applying (A16) and (A12) on its actual quotient/subspace space yields

$$\|h\|_4^4 \leq 2^{114d^2}B\|h\|_2^2 \leq 2^{196d^2}\epsilon^2. \quad (\text{A20})$$

Both factors B are required. Since raw restriction commutes with squaring and $\deg(f^2) \leq 2d$, the actual square is up-to- $2d$ global with parameter $2^{196d^2}\epsilon^2$.

For every dyadic $p \geq 2$ we claim

$$\|f\|_p^p \leq 2^{200d^2p^2}\|f\|_2^2\epsilon^{p/2-1}. \quad (\text{A21})$$

At $p = 2$ the last factor is one. If $\epsilon = 0$ then $f = 0$; if $d = 0$ then $f = c$ is constant and $|c|^2 \leq \epsilon$, proving the claim without loss. For $d \geq 1$, $\epsilon > 0$, start at $p = 4$ with (A19). Define $A_4 = 114$ and $A_p = 4A_{p/2} + 49p - 82$ for $p \geq 8$. Inductively apply the stronger moment bound with exponent $A_{p/2}$ to $g = f^2$, its actual degree bound $2d$, and the parameter from (A20). Using $\|g\|_2^2 = \|f\|_4^4$ and (A19), the total exponent is

$$4A_{p/2} + 196(p/4 - 1) + 114 = 4A_{p/2} + 49p - 82,$$

and the density power is $2(p/4 - 1) + 1 = p/2 - 1$. The factor four retains the doubled degree. The invariant $A_p \leq 200p^2 - 100p$ holds at $p = 4$ and propagates because $4(200(p/2)^2 - 100(p/2)) + 49p - 82 = 200p^2 - 151p - 82$. This proves (A21) for $2, 4, 8, \dots$ on all finite dimensions. No exact-order globalness is inferred on an undersized quotient, and no dyadic theorem was imported to prove (A21).

A.7 The dual-norm level bridge and the Boolean conclusion

Let $p \geq 2$ be dyadic and $p' = p/(p-1)$. For arbitrary f , suppose every restriction through order i has $L^{p'}$ norm at most ϵ . This is a norm, not a squared norm. We prove

$$I_{A,B,T}(f^{\neg i}) \leq 2^{500i^2p} \epsilon^2 \quad \text{for } |(A,B)| \leq i. \quad (\text{A22})$$

For $p = 2$ use (A15). For $p \geq 4$ the following duality estimate will be used only after its influence premise has been established. Put $h = f^{\neg j}$, $E = \|h\|_2^2 > 0$, and suppose all its influences through j are at most βE (necessarily $\beta \geq 1$). By (A18) its globalness parameter is $2^{10j^2} \beta E$. Taking the p th root in (A21) gives

$$\|h\|_p \leq 2^{210j^2p} \beta^{1/2-1/p} \sqrt{E}.$$

Since the actual orthogonal projection has $|\langle h, f \rangle| = E$, Holder and division by $\sqrt{E}$ give

$$E \leq 2^{420j^2p} \beta^{1-2/p} \|f\|_{p'}^2. \quad (\text{A23})$$

The zero-energy case is direct.

Induct on j for (A22), simultaneously on all finite spaces. At $j = 0$ the level is constant and Holder gives $|\mathbb{E}f| \leq \epsilon$. The translation mixtures E_U, E_U^2 also preserve globalness in $L^{p'}$ by Minkowski. The polynomial in (A14) has coefficient sum at most 2^{3j} for $j \geq 1$. Thus its explicit witness at every base is up-to- $(j-1)$ global in $L^{p'}$ with parameter $2^{3j}\epsilon$. The inductive hypothesis and (A14) bound every positive-order influence of $f^{\neg j}$ by

$$B = 2^{500(j-1)^2p+6j} \epsilon^2.$$

All positive orders factor through an order-one derivative as in (A15), so none is omitted. Let $E = \|f^{\neg j}\|_2^2$. If $E \leq B$, every influence is at most B , which is at most the bound in (A22) since $6j \leq 500(2j-1)p$. If $E > B$, the proved positive-order bounds and the exact zero-order energy show that every influence is at most E . Only now apply (A23) with $\beta = 1$, obtaining $E \leq 2^{420j^2p} \epsilon^2$ and hence (A22). There is no assumption of the desired influence bound in this case split, and (A21) was proved without using this bridge.

Proof of Theorem 2. Every actual affine restriction $T + \mathcal{H}(\mathbb{F}_2^D/A, B)$ of order at most $i \leq r$ is a solution set of vector equations: choose a basis of A to fix M on it as T does, and quotient rows for B to impose the codomain constraints. Pad with zero-equals-zero equations to make the nominal budget r . Neither the solution set nor its uniform measure changes. Thus all these restrictions have density at most δ , including the padded empty restriction, which gives $\mathbb{E}F \leq \delta$. For a Boolean function, its $L^{p'}$ norm on a restriction of density a is $a^{1/p'}$. Apply (A22) with $\epsilon = \delta^{1-1/p}$ to $h = F^{\neg i}$. For $i \geq 1$, $\delta > 0$ its influence parameter is $\eta = 2^{500i^2p} \delta^{2-2/p}$. By (A18) its up-to- i squared- L_2 globalness parameter is $\eta_1 = 2^{(10+500p)i^2} \delta^{2-2/p}$. Apply (A21) and use Parseval and Booleanity: $\|h\|_2^2 \leq \|F\|_2^2 = \mathbb{E}F \leq \delta$. Hence

$$\begin{aligned} \|h\|_p^p &\leq 2^{(450p^2-495p-10)i^2} \delta^{p-2+2/p} \\ &\leq 2^{500i^2p^2} \delta^{p-2}. \end{aligned}$$

The calculations are $200p^2 + (10+500p)(p/2-1) = 450p^2 - 495p - 10$ and $1 + (2-2/p)(p/2-1) = p-2+2/p$. The last inequality uses $0 < \delta \leq 1$ in the correct direction. Take the p th root. If $\delta = 0$, the padded empty restriction forces $F = 0$. At $i = 0$, the projection is the constant $\mathbb{E}F \leq \delta \leq \delta^{1-2/p}$; unavailable ranks give zero. These cases also cover zero-dimensional matrices. $\square$

If one instead starts with independent restrictions of order exactly i , an up-to- i bound follows when $i \leq \dim V + \dim W$ by partitioning each lower-order affine variation space into cosets of a chosen order- i variation subspace and averaging the relevant norm powers. For a rank- i conclusion, larger i gives the zero level directly. For a full bounded-degree function, a vacuous exact-order premise does *not* give globalness; all full-function uses above explicitly assume up-to-order bounds. The nominal-budget hypothesis of Theorem 2 avoids this issue by zero padding. In Lemma 5 take $\delta = 2e \leq 1$ and the stated dyadic moment. Probability-norm monotonicity supplies any smaller moment used there. This discharges precisely the binary analytic input, without changing the remaining upstream contracts of the manuscript.

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